

Automated Tumor Name Standardization in the NIH Clinical Trials Registry Using the CANTOS Pipeline

Aditya Lahiri¹, Sangeeta Shukla¹, Ben Stear¹, Taha Mohseni Ahooyi¹, Katherine Beigel¹, Elizabeth Margolskee², Deanne Taylor^{1,3}

Affiliations:

1. The Department of Biomedical and Health Informatics, The Children's Hospital of Philadelphia, Philadelphia PA
2. Department of Pathology & Laboratory Medicine, Children's Hospital of Philadelphia, Philadelphia PA
3. Department of Pediatrics, University of Pennsylvania Perelman Medical School, Philadelphia PA

The bigger picture

The National Institutes of Health's (NIH) clinical trials registry (CTR, ClinicalTrials.gov) is a public database that provides a wide range of information regarding a clinical trial, such as outcomes, interventions, conditions studied, and experimental design. This information is distributed in various text files each containing the National Clinical Trial Identifier number (NCT ID) to identify and aggregate data for a given clinical trial. In pursuit of studying the landscape of therapeutic agents and drug targets associated with cancer types and tumor names, the CTR serves as a valuable resource. However, the tumor names contained in the "conditions" file provided at the CTR are often unstandardized and thus difficult to identify using simple text matching, creating a barrier to further analysis. Here we present an and compare methods to identify and standardize tumor names from the CTR in order to facilitate downstream analysis and data integration.

Summary

This study aimed to identify and standardize tumor names from the NIH's clinical trials registry (CTR, ClinicalTrials.gov) using the World Health Organization Classification of Tumours (WHO system) and the National Cancer Institute Thesaurus (NCIt). We developed CANTOS (Clinical trials Automated Nomenclature and Tumor Ontology Standardization), a computational pipeline that identified tumor names from the CTR and standardized them against the WHO system and NCIt database. We evaluated several methods based on text matching and text-embedding to standardize the tumor names from the CTR and evaluated the accuracy of these methods only against the WHO system, as it is considered a clinical standard for tumor nomenclature. Text-embeddings were generated for tumor names in the CTR, WHO system, and NCIt database using several open source and proprietary models and provided as input to CANTOS. Our study demonstrated that several text-embedding based methods significantly outperformed text-matching approaches in the standardizing CTR tumor names. However, depending on the text-embedding, the standardization accuracy of the text-embedding methods varied significantly when standardizing against the WHO 5th edition (latest) and WHO all editions (3rd–5th). Notably, the text-embedding based method LTE-3+Euclidean Distance, achieved the highest accuracy of 69.1% when standardized against WHO system all editions, whereas the method all-MiniLM-L12-v2+Euclidean Dist achieved the highest standardization accuracy of 67.8% against the WHO system 5th edition.

Keywords

NIH Clinical Trials Registry; WHO Tumor Classification; Text Embedding; Clinical Text Standardization; Large Language Models

Introduction

Cancer is a significant global health issue and the second leading cause of death in the United States ¹. In 2024, over 2 million new cancer cases and 611,720 deaths are expected in the U.S. alone ². While pediatric cancer survival rates have improved to 80% over the last five decades ^{3,4}, which can primarily be attributed to successes in the treatment of common childhood hematological malignancies such as acute lymphoblastic leukemia ⁵, unfortunately this success is not uniformly shared ⁶, and certain cancers, particularly those of the brain and nervous system ², remain difficult to treat. Pediatric cancers are rarer than adult cancers and face unique challenges in clinical trials, including limited therapeutic agents, difficulty recruiting diverse populations, and tumor heterogeneity ^{7,8}. To understand the therapeutic landscape of adult and pediatric tumors, it is critical to extract and analyze data from various biomedical databases, particularly from the NIH's Clinical Trials Registry (ClinicalTrials.gov, CTR) which contains data from over 482,529 research studies across all 50 states in the US and 223 countries ⁹. While the CTR has established protocols and guidelines for data the submission process, tumor names submitted to the CTR (provided by CTR in their "conditions.txt" file) contain extraneous information, typographical errors, missing values, non-standard nomenclature and other inconsistencies which create barriers for data integration and downstream analysis of the CTR data. Although the CTR mandates standardized terminology like Medical Subject Headings (MeSH) ¹⁰, these terms often fail to capture the specific details of many tumor types outlined in the conditions data file. In **Supplementary Document SD1**, we provide examples to compare the differences between the terms in the condition file and the MeSH terms. Thus there is a need to standardize the CTR conditions data file. Ideally, tumor names from the CTR would be classified using ontologies and/or standards, including from the WHO Classification of Tumours ¹¹⁻⁴⁴ system (we refer to here as the "WHO system", <https://tumourclassification.iarc.who.int/welcome/>) or National Cancer Institute Thesaurus (we refer to as the "NCIt database" <https://ncithesaurus.nci.nih.gov/ncitbrowser/>). The WHO and NCIt standardized nomenclature are available in **Supplementary Tables ST1-ST3**.

To analyze CTR data with accurately classified pediatric and adult tumor types, it is essential to standardize tumor names using controlled vocabularies from the WHO Classification of Tumours (WHO system) and the National Cancer Institute Thesaurus (NCIt database). To address this challenge, we developed CANTOS (Clinical Trials Automated Nomenclature and Tumor Ontology Standardization), a computational pipeline that standardizes tumor names in the CTR's "conditions" file. By systematically mapping tumor names to their standardized counterparts in the WHO and NCIt databases, CANTOS enables seamless integration of CTR data with external biomedical resources such as Open Targets (OT, <https://www.opentargets.org/>)⁴⁵ and Illuminating the Druggable Genome (IDG, <https://druggablegenome.net/>)⁴⁶, facilitating a better and more comprehensive understanding of tumor biology, drug-target interactions, and therapeutic agents.

CANTOS employs text-matching and text-embedding approaches to standardize tumor names. Text-matching methods, including Levenshtein distance, Jaro-Winkler distance, and cosine similarity,

compare tumor names at the character or token level. While these methods efficiently identify minor variations, they struggle to resolve semantic differences between non-identical but clinically synonymous terms.

To improve standardization accuracy, we incorporated text-embedding approaches, which transform tumor names into high-dimensional vector representations, enabling semantic comparisons. In this study, embeddings were generated using 26 different transformer-based models and large language models (LLMs), including both open-source and proprietary models. Open-source models include e5-large^{47,48}, e5-large-v2^{47,49}, all-mpnet-base-v2⁵⁰, all-MiniLM-L12-v2⁵¹, gtr-t5-large^{52,53}, all-MiniLM-L6-v2⁵⁴, all-roberta-large-v1⁵⁵, sapBERT^{56,57}, clinicalBERT^{58,59}, PubMedBERT^{60,61}, BioBERT⁶², SciBERT^{63,64}, ModernBERT^{65,66}, BioGPT^{67,68}, MedLlama2^{69,70}, MedLlama_13B⁷¹, LaBSE^{72,73}, Llama 3.3^{74,75} (Llama3.3_70B), Llama3.2^{75,76} (Llama3.2_3B), Llama3.0^{75,77}, Phi-4^{78,79} nomic⁸⁰, and DeepSeek_8B^{81,82}, many of which were obtained from Hugging Face (<https://huggingface.co/>) or Ollama (<https://ollama.com/>) interfaces. Some of these models such as BioBERT, SciBERT, BioGPT, clinicalBERT, PubMedBERT, MedLlama2, and MedLlama_13B were trained on biomedical, clinical and scientific texts. Proprietary models tested include OpenAI's text-embedding-ada-002 (ADA-002)⁸³ and text-embedding-3-large (LTE-3)⁸⁴ and Cohere's embed-english-v2.0⁸⁵, which were accessed via API and incur costs based on computational resource consumption. Unlike open-source models, proprietary embeddings offer no transparency regarding architecture or weights, limiting their adaptability for fine-tuning.

A key motivation for this study was to assess how well general-purpose transformer-based models and LLMs perform in biomedical nomenclature standardization tasks without additional fine-tuning. While some models, such as BioBERT and PubMedBERT, have been pre-trained on biomedical corpora, most models evaluated in this study were not specifically optimized for biomedical terminology tasks. Given the increasing use of LLM and transformer-based embeddings in text standardization^{86–90}, we sought to determine whether general-purpose and domain-adapted models can effectively align CTR tumor names with WHO classifications without further model-specific adaptation.

To systematically compare these models, CANTOS employs strategies such as pairwise Euclidean distance calculations between tumor names extracted from the CTR, WHO system, and NCIt database. For each tumor name in the CTR, CANTOS identifies the nearest standardized term in the WHO 5th edition, WHO all editions dataset, and the NCIt database, mapping each tumor entry to its closest standardized counterpart. This approach allows for a structured evaluation of different embedding methods in clinical text standardization.

Beyond pairwise distance-based standardization, CANTOS applies clustering methods, including K-means and Affinity Propagation (AP), to identify patterns in tumor nomenclature and reduce ambiguity in term assignments. Additionally, majority voting is used as an ensemble approach,

aggregating results from multiple high-performing embedding models to enhance standardization accuracy.

By systematically evaluating diverse text-matching, embedding, and clustering strategies, this study provides a comprehensive benchmark for clinical text standardization. While this work focuses on tumor nomenclature, CANTOS is a generalizable framework that can be applied to other biomedical standardization tasks, including harmonizing disease terminology across ontologies such as MONDO, Disease Ontology (DO), Human Phenotype Ontology (HPO), ICD-10, and SNOMED CT. Future work will explore the integration of biomedical-specific LLMs fine-tuned on tumor terminology, further enhancing the accuracy and applicability of CANTOS for exploring performance of other large-scale clinical data harmonization tasks.

Results

In this paper, we present the CANTOS pipeline, which extracts tumor names from the CTR and standardizes them using the WHO system and NCIt database. We evaluate its standardization performance across different methods.

CANTOS extracted 13,230 tumor records from the CTR, with 6,324 identified as pediatric, and then implemented 36 single or combined methods based on text-matching and text-embedding to standardize the tumor names from the CTR. To evaluate these methods, we needed ground truth (i.e., standardized terms) for all the 13,230 tumor names. In order to provide a ground-truth set for testing, we randomly sampled 1,600 tumor names from the 13,230 and manually annotated those tumor names to assess our methods' performance.

For these 1,600 tumor names, we annotated two “ground truth” definitions from the WHO system based on release versions: the “WHO 5th edition”^{11–20} annotation based on the latest system release, and the “WHO all editions” which combines annotations from the WHO system’s 3rd, 4th, and 5th editions^{11–44}. We used the WHO system (and not NCIt), exclusively as our annotation standard because the WHO system is considered a clinical standard for tumor nomenclature. During annotation, we encountered some CTR tumor names that could have multiple ground truth matches or no matches within the WHO system. CTR tumor names with multiple ground truth matches were included in the accuracy evaluation, while those without any ground truth matches were excluded. When standardizing using the WHO 5th edition, we identified 556 CTR tumor names that did not have a ground truth, while 482 CTR tumor names did not have a ground truth when WHO all editions were used. This difference arises because the WHO all editions collectively contain more standardized tumor terms than the 5th edition. Consequently, we evaluated the standardization accuracy of each method against 1,044 terms when using the WHO 5th edition and 1,118 terms for the WHO all editions set. Standardization accuracy for each method was calculated by counting instances where a method identified at least one ground truth associated with a CTR tumor name and dividing by the total number of terms (1,044 or

1,118). We report standardization accuracy for WHO all editions in **Table 1** and the 5th edition in **Table 2**. For clarity in discussion, we categorized the standardization methods based on their standardization accuracy as follows: high-accuracy methods (standardization accuracy $\geq 60\%$), intermediate-accuracy methods ($50\% \leq \text{standardization accuracy} < 60\%$), and low-accuracy methods (standardization accuracy $< 50\%$).

Tables 1 and **2** show that several text-embedding-based methods have equally high performance and achieve a comparatively higher standardization accuracy, irrespective of the edition of the WHO system used for standardization. However, text-embedding methods based on models such as PubMedBERT, BioBERT, SciBERT, ModernBERT, MedLlama2, MedLlama_13B, Llama3.3_70B, Llama3.2_3B, Llama3.0, Phi-4, DeepSeek_8B have comparable or poorer standardization accuracy compared to text-matching methods. The text-embedding method LTE-3+Euclidean Dist achieved the highest accuracy of 69.3% when standardizing the CTR tumor names against WHO all edition, it was followed by the method e5-large+Euclidean Dist which achieved a standardization accuracy of 68.2% (see **Table 1**). Several other methods based on text-embedding derived from LTE-3, ADA-002, all-mpnet-base-v2, all-MiniLM-L12-v2, all-MiniLM-L6-v2 and nomic, achieve standardization accuracies above the 60% mark when standardizing against the WHO all editions set. In **Table 2**, when standardizing against the WHO 5th edition, we observe that the methods all-MiniLM-L12-v2+Euclidean Dist achieves the highest standardization accuracy of 67.8%, however, this method has only a marginally higher standardization accuracy compared to methods such as e5-large+Euclidean Dist and all-mpnet-base-v2+Euclidean Dist which have a standardization accuracy 67.4% and 67.1% respectively. Other methods based on text-embedding from ADA-002, LTE-3, all-MiniLM-L6-v2, and nomic achieved a standardization accuracy above 60% against the WHO 5th edition. From **Tables 1** and **2**, it can be noted that methods using text-embedding derived from an updated or more powerful version of their base text-embedding model had a higher standardization accuracy. For example, consider the methods utilizing text-embedding derived from Open AI's proprietary models text-embedding-ada-002 (ADA-002) and text-embedding-3-large (LTE-3) in **Tables 1** and **2**. LTE-3 is a more recent and larger model producing text-embedding of higher dimensions than ADA-002. In general, LTE-3 text-embedding performed better than ADA-002, except for the LTE-3+K-means method, which slightly underperformed compared to ADA-002+AP(Affinity Propagation) and ADA-002+Euclidean Dist when using WHO all editions (**Table 1**). However, LTE-3+K-means ranked higher than its ADA-002 variant method, ADA-002+K-means, for both editions of the WHO system. Similarly, we notice in both **Tables 1** and **2** that methods based on all-MiniLM-L12-v2, MedLlama_13B, and Llama3.3_70B performs better than methods based on their previous or smaller size models such as all-MiniLM-L6-v2, MedLlama2, and Llama3.2_3B, Llama3.0 respectively. The performance variation in text-embedding models can be attributed to various factors that are involved in the generation of the embeddings such as the model architecture, training data, domain specialization, and embedding dimensions. The only exception to this observation is when we consider the methods e5-large+Euclidean Dist and e5-large-v2+Euclidean dist, where the standardization accuracy dropped significantly across WHO 5th and WHO all editions when using text-embedding obtained from the updated model e5-large-v2. Also when standardizing against

the WHO 5th edition, the methods all-MiniLM-L6-v2+Euclidean Dist and all-MiniLM-L12-v2+Euclidean Dist were tied with the same standardization accuracy, with all-MiniLM-L12-v2 being the larger model.

When the CANTOS pipeline standardizes the tumor names from the CTR against the WHO all editions, the method LTE-3+Euclidean Dist achieved the highest accuracy, whereas the method all-MiniLM-L12-v2+Euclidean Dist had the highest accuracies when standardizing against WHO 5th edition. In **Figures 1** and **2**, we visualize the standardization performance of these methods. In each of these figures, for the given embedding space, we display the distribution of the Euclidean distances between the CTR tumor names and the closest (minimum Euclidean distance) standardized tumor name from the WHO system to which it was mapped. It should be noted that out of the 1600 randomly sampled CTR tumor names that we manually validated, only the terms which had a ground truth from the WHO 5th or WHO all editions were considered for evaluation of standardization accuracy and are displayed in **Figure 1** and **2**. Thus, in **Figures 1a** and **2a** we have the closest Euclidean distance distribution of 1,044 CTR tumor names for which ground truths were identified in the WHO 5th edition and in **Figure 1b** and **2b** we have the closest Euclidean distance distribution of 1,118 CTR tumor names for which ground truths were identified in the WHO all editions. For both the LTE-3+Euclidean Distance and all-MiniLM-L12-v2+Euclidean Distance methods, the median Euclidean distance for correctly standardized CTR tumor names was consistently lower than that for incorrectly standardized names across both the WHO 5th edition and WHO all editions. This finding is expected, as correctly mapped CTR tumor names tend to be more syntactically and semantically similar to their corresponding standardized WHO terms. As a result, their Euclidean distances are lower compared to cases where the mappings were incorrect.

Tables 3 and **4** present the summary statistics for Euclidean distances presented in **Figures 1** and **2**, respectively. These statistics may offer valuable insights for establishing thresholds within CANTOS to assess the reliability of the standardization methods. For instance, consider the method LTE-3+Euclidean Distance when the WHO 5th edition is used in CANTOS (see **Table 3**), the median Euclidean distance for correctly standardized terms is 0.696, while for incorrectly standardized terms it is 0.8408. If 0.696 was adopted as the reliability threshold, then any CTR tumor name mapped to a standardized WHO term with a distance below this value would be considered a reliable standardization. However, as indicated in **Table 3**, the minimum Euclidean distance at which a CTR tumor name was incorrectly standardized is 0.339. This implies that adopting 0.696 as the reliability threshold would incorrectly designate some tumor names as reliably standardized by this method. Conversely, using 0.339 as the reliability threshold would misclassify several correctly standardized tumor names as unreliable standardization. This finding highlights the limitations of thresholds based solely on Euclidean distance distributions for reliably distinguishing between correct and incorrect standardizations in CANTOS. The lack of clear separation between these distributions, as shown in **Figure 1**, underscores the need for higher-quality text embeddings that would yield greater class distribution differences between correctly and incorrectly standardized tumor names. Such embeddings

would also reduce the number of incorrect standardizations, thereby improving standardization accuracy. LLMs or transformer based models trained and fine-tuned on extensive, diverse datasets, including tumor nomenclature from the WHO system and the NCIt database are likely to be promising tools for generating such high-quality embeddings.

Among the text-matching methods based on edit distances, Levenshtein distance outperformed both Jarro-Winkler and cosine distances or any of their implementations using AP clustering. Jarro-Winkler is effective for minor text discrepancies and common prefixes, while cosine distance relies on word frequency ("bag of words") and ignores word order. In contrast, Levenshtein distance counts the minimum edit operations needed to transform one text into another while maintaining word order and without considering prefix similarity, which likely contributed to its better performance. It should be noted that the text-matching methods had better or similar performance to various text-embedding methods based on models such as Llama3.3_70B, Llama3.2_3B, Llama3.0, MedLlama2, MedLlama_13B, BioBERT, SciBERT, PubMedBERT, ModernBERT, Phi-4 and DeepSeek_8B. Compared to text-embeddings, text-matching methods are computationally cheaper to run and are straightforward methods that are not based on complex models. Thus, we conclude that not all text-embeddings are appropriate for the task of standardization of tumor names and reinforces the need for models that are trained and fine tuned on standardized tumor nomenclature.

We also extended our analysis to combine three high-accuracy standardization methods (out of eleven high-accuracy methods, see **Tables 1** and **2**) and selected the combination of methods that had the lowest average pairwise mutual information score to maximize the diversity in their standardization prediction and thereby increase standardization accuracy. The combined methods predicted the standardization for a given CTR tumor name by performing a majority vote operation. This strategy led to the combination of the methods:ADA-002+AP, LTE-3+K-means, and all-MiniLM-L12-v2+Euclidean Dist, for the WHO 5th edition which achieved a standardization accuracy of 72.1%. Whereas for WHO all editions, the methods that were combined included:ADA-002+K-means, LTE-3+K-means, and all-MiniLM-L12-v2+Euclidean Dist, which achieved an accuracy of 71.4%. In addition to this mutual information-based majority voting standardization method, we also evaluated the standardization accuracy when all eleven high-accuracy standardization methods (see **Tables 1** and **2**) were used to perform the majority vote and predict the standardized tumor name for a given CTR tumor name. We observed that incorporating all eleven high-accuracy methods resulted in a decline in standardization accuracy, yielding 44.2% for the WHO 5th edition and 48.4% for WHO all editions.

Ground truth annotations for the 1,600 CTR tumor names and their standardization results are available in **Supplementary Tables ST4** (WHO all editions) and **ST5** (WHO 5th edition). Standardized terms for each tumor in the CTR are reported in **Supplementary Tables ST6-ST9**.

Discussion

The CTR provides information that can be integrated with external databases to gain insights into therapeutic and drug-target landscapes for tumors. However, raw tumor names from the CTR cannot be directly linked to other databases due to inconsistencies such as extraneous information, typographical errors, missing values, and non-standard nomenclature. Since tumor names are not automatically identifiable in the CTR, they also need to be extracted. To address this, we developed CANTOS, a computational pipeline to extract and standardize tumor names from the CTR according to the WHO system and NCIt database.

CANTOS classified tumor names from the CTR as adult or pediatric, and we manually validated each classification. For pediatric tumors, we assigned citations that confirmed the classification (**Supplementary Table ST10**). CANTOS identified 13,230 unique tumors, of which 6,324 were pediatric tumors. It standardized these names according to the WHO system and NCIt database using 36 methods based on text-matching (edit distances) and text-embedding. We evaluated the standardization accuracies of these methods only against the WHO system, which is considered the gold standard for clinical tumor nomenclature. To assess standardization accuracy, we randomly selected 1,600 tumor names from the 13,230 standardized by CANTOS and manually annotated their ground truths using the WHO 5th edition and WHO all editions. More tumors were identifiable using WHO all editions, as the WHO 5th edition contains fewer terms due to nomenclature updates, and older tumor names recorded in the CTR decades ago were more likely to match entries in WHO all editions.

Among the methods, text-matching based approaches achieved standardization accuracy ranging from 21.6% to 30.3% against the WHO 5th edition and from 22.7% to 32.6% against WHO all editions. In contrast, text-embedding-based methods exhibited a wider accuracy range, from 13.7% to 67.8% for the WHO 5th edition and from 15.4% to 69.1% for WHO all editions. Unlike the text-matching based methods which only capture the syntactical difference between the unstandardized CTR tumor name and standardized tumor name from the WHO system, text-embedding are designed to capture both syntactic and semantic information in a text, which is likely why some text-embedding based methods achieved high standardization accuracy. However, not all text-embedding methods had higher accuracy than text-matching methods, depending on the model from which the text-embeddings were generated, the standardization accuracy varied significantly. Although the variance in standardization accuracy is broader for text-embedding methods, their performance (i.e. rankings, see **Tables 1** and **2**) remained consistent across both the WHO 5th and WHO all editions.

When standardizing against either the WHO 5th or WHO all editions, methods based on text-embeddings generated from the models LTE-3, ADA-002, all-MiniLM-L12-v2, all-MiniLM-L6-v2, e5-large, all-mpnet-base-v2, and nomic achieved a minimum standardization accuracy of at least 60.2% and were consequently classified as high-accuracy methods. The highest standardization accuracy is obtained by the methods LTE-3+Euclidean Dist when standardizing against the WHO all editions and by all-MiniLM-L12-v2+Euclidean Dist when standardizing against the WHO 5th edition. Methods based on text-embeddings generated from all-roberta-large-v1, gtr-t5-large, embed-english-v2.0, and

e5-large-v2 are classified as intermediate-accuracy methods for both WHO 5th and WHO all editions. SapBERT was classified as both an intermediate and low-accuracy method as its standardization accuracy was 51.7% against the WHO 5th editions and 47.9% against the WHO all editions. The low-accuracy text-embedding methods were based on models such as Llama3.3_70B, Llama3.2_3B, Llama3.0, MedLlama2, MedLlama_13B, BioBERT, SciBERT, PubMedBERT, ModernBERT, Phi-4 and DeepSeek_8B. The standardization accuracies of the low-accuracy models were either comparable or poorer compared to the text-matching methods. All text-matching based methods were classified as low-accuracy methods.

For all the text-embedding based methods, we standardized the CTR tumor terms by identifying the nearest standardized term (closest match method) from the WHO system or NCIt database in the respective text-embedding space using Euclidean distance. In addition to the closest match method, We also implemented a standardization method that combined text-embedding obtained from OpenAI's models (LTE-3 and ADA-002) with Affinity Propagation clustering (LTE-3+AP, ADA-002+AP) and K-means (LTE-3+K-means, ADA-002+K-means) clustering. We limited clustering based approach to use Open AI's text-embedding models LTE-3 and ADA-002 as they achieved high and consistent standardization accuracy against both the WHO 5th and WHO all editions. Furthermore, combining clustering with text-embeddings makes these methods computationally expensive, which prompted us to limit these methods to use LTE-3 and ADA-002 text-embeddings. We observed that methods combining text-embeddings and clustering had comparable but lower standardizing accuracy as compared to their counterpart closest match methods (LTE-3+ Euclidean Dist , ADA-002+ Euclidean Dist). In general, the LTE-3 based methods generally outperformed ADA-002, except for LTE-3+K-means, which marginally underperformed compared to ADA-002+Euclidean Dist and ADA-002+AP when standardized against WHO all editions. However, LTE-3+K-means performed better than its counterpart method ADA-002+K-means against either editions of the WHO system. Compared to these methods that combine text-embedding and clustering, the text-embedding based methods using closest match approach, achieve higher standardization accuracy and are simpler and faster as they do not require additional steps like clustering, cluster size analysis, or outlier detection.

In our effort to improve the standardization accuracy, we devised a method that standardized a given CTR tumor name by performing a majority vote on the standardization predictions obtained from a combination of high-accuracy methods. We considered combinations of three methods as an odd number of methods reduces chances of a tie occurring. We limited the majority vote to only the high-accuracy methods identified in **Tables 1** and **2**. Since eleven methods were classified as high-accuracy methods for both the WHO 5th edition and WHO all editions, it implied that there were a total 165 ($^{11}C_3$) possible three-method combinations for either editions of the WHO system. Instead of testing each combination of methods, we selected three methods based on the lowest average pairwise mutual information. This allowed for selecting methods that had high levels of disagreement (i.e., high diversity in their predictions) in their prediction, thereby increasing the chances of one method predicting the standardization correctly when another method failed. Using this strategy we identified the combination

of methods ADA-002+AP, LTE-3+K-means, and all-MiniLM-L12-v2+Euclidean Dist, which had the lowest average pairwise mutual information for the WHO 5th edition and achieved a standardization accuracy of 72.1%. Whereas the combination of methods ADA-002+K-means, LTE-3+K-means, and all-MiniLM-L12-v2+Euclidean Dist, which had the lowest average pairwise mutual information for the WHO all editions and achieved a standardization accuracy of 71.4%. Compared to the methods such as all-MiniLM-L12-v2+Euclidean Dist and LTE-3+Euclidean Dist which achieved a highest standardization accuracy for the WHO 5th edition and WHO all editions (respectively), the majority vote method improved by ~4% (~41 more terms were correctly standardized) for the WHO 5th edition and by ~2% (~22 more terms were correctly standardized) for WHO all editions. It should be noted that this improvement is with respect to the subset of the 1600 tumor names sampled from the CTR for which a ground truth could be identified in the WHO system. While the standardization accuracy of the selected combination of methods for majority vote improved for both the WHO 5th and WHO all editions for this subset of CTR tumor names, this improvement may not generalize to a different subset of tumor names from the CTR. Furthermore, we also tested the standardization accuracies when all the eleven methods were used for performing the majority vote to predict the standardization for a given CTR tumor name. This method led to a sharp decline in the standardization accuracy across both the WHO 5th and WHO all editions thus highlighting the advantage of using mutual information to selectively choose a subset of methods for majority voting, which contributed towards improved standardization accuracy.

The CANTOS pipeline is adaptable for standardizing biomedical nomenclature across various systems beyond the WHO system and NCIt databases. It can be extended to work with other ontologies such as MONDO⁹¹, Disease Ontology (DO)^{92–102}, Human Phenotype Ontology (HPO)¹⁰³, and medical coding systems like ICD-10¹⁰⁴ and SNOMED¹⁰⁵. Future work includes generalizing CANTOS to incorporate additional embedding methods, including domain-specific biomedical LLMs or **transformer based models**, and expanding its functionality to standardize any collection of biomedical terms.

Limitations of the study

As more CTR samples are annotated with ground truth, the standardization accuracies of each method will approach their true values. While we expect accuracy changes, our annotation of 1,600 CTR tumor names suggests that text-embedding methods will likely outperform text-matching methods. Expert annotation of CTR tumor names is crucial for accurately evaluating these methods and is a limitation of our study. Additionally, as the CTR updates, new tumor names must be identified and standardized, necessitating rerunning CANTOS and manually validating the findings, which is time-consuming. Beyond expert annotation, there are computational costs associated with running the CANTOS pipeline, generating text-embedding, and storing data, which can become expensive over time. Another limitation is that text-embedding generated using the current versions of the LLMs or transformer based models need to be updated as newer and more powerful models become available. While we tested methods that employed text-embeddings from models that were trained on biomedical and scientific literature such as MedLlama, PubMedBERT, SciBERT, BioGPT, and ClinicalBERT, these models did not achieve high

standardization accuracy. Therefore, it is not necessarily true that a domain-specific model will produce high quality text-embeddings sufficient for disambiguation of tumor nomenclature. Even though these models are trained on biomedical literature, their poor performance can possibly be attributed to low generalization power, which can happen when the training dataset is not rich or diverse enough. Furthermore, having a very focused and narrow training dataset can impede the performance of text-embedding due to limitations on the deeper contextual understanding of words. Contextual understanding can potentially help text-embedding to disambiguate terms and detect nuanced relationships between words. Various other factors such as model architecture and embedding dimensions also play a role in the quality of text-embeddings produced. Any LLM or transformer based model that is trained and fine tuned on a large diverse dataset that include tumor nomenclature will be a potential candidate for future integration in the CANTOS pipeline as text-embeddings generated from these models could likely differentiate tumor names more precisely¹⁰⁶.

The CANTOS pipeline uses text-embeddings from LLMs and transformer based models that may inadvertently produce outputs reflecting biases inherent in their training data. It is important to recognize that LLMs and transformer based models outputs are not always neutral or free from error, and results should be interpreted with caution. Importantly, the CANTOS pipeline is designed solely for research analysis purposes and should not be applied in clinical or diagnostic settings. This tool is not intended for use in contexts where decisions may directly affect human health, treatment, or care. Any conclusions drawn from the use of CANTOS in research should be further reviewed and validated by clinical professionals and subjected to rigorous peer-reviewed testing before any potential medical or therapeutic applications.

The CANTOS pipeline facilitates the extraction and standardization of tumor names from the CTR using several methods. While guidelines exist for submitting data to the CTR to ensure basic integrity, there are no enforced protocols for standardizing tumor names in the conditions file. Although studies have emphasized the need for standardization in clinical trials regarding study design^{107,108} and evidence reproducibility¹⁰⁹, none have focused on standardizing tumors or condition names in the CTR. Standardizing tumor names will enhance their searchability in other biomedical databases, enabling researchers to quickly gather information on associated therapeutic agents, drug-targets, and clinical outcomes for specific tumors.

Experimental procedures

This section outlines the design and methods of the CANTOS computational pipeline, which performs two main tasks: (i) identifying tumor names from the CTR and (ii) standardizing CTR tumor names. CANTOS takes condition names from CTR and standardized tumor names from the WHO system and NCIt database as inputs. After identifying tumor names, CANTOS iterates through the standardization step twice to account for different editions of the WHO system. The WHO system is considered the gold standard for tumor nomenclature and has multiple versions due to updates over the years. We used the 3rd, 4th, and 5th editions of the WHO system, which are all publicly available online. In the first

iteration, CANTOS used only the latest 5th edition of the WHO system, while the second iteration incorporated all editions of the WHO system. In other words, this approach first aims to standardize the tumor names from the CTR based on only the latest edition of the WHO system, i.e., 5th edition and then follows by standardizing the tumor names in CTR with the combined editions of the WHO system, i.e., WHO system all editions (includes 3rd, 4th, and 5th referred to in this text as the “WHO all edition”). Throughout these two iterations, tumor names from the CTR were also standardized against the NCI database.

Identification of tumor names from CTR

The CTR provides information on various aspects of clinical trials, such as outcomes, interventions, conditions, experimental design, and study sponsors. The CTR data is distributed across multiple text files, and each of these files contains the National Clinical Trial Identification Number (NCTID). The NCTID serves as a unique identifier for each trial, which allows for referencing and aggregating related information distributed across the files. In this study, we focused on the conditions ("conditions.txt" or conditions file) and interventions ("intervention.txt" or interventions file), which detail the conditions and drugs used in each trial, respectively. The conditions file contains 801,197 records and is annotated with the fields: "id", "nct_id", "name", and "downcase_name". The "id" represents the record number within the conditions file, while the "nct_id" is the foreign key. The "name" and "downcase_name" fields include the names of the studied conditions, with "downcase_name" in lowercase format. We identified 105,483 unique conditions based on string uniqueness in the conditions file.

The conditions file does not categorize the condition names into specific subtypes or classes, consequently the tumor names contained in the file are not easily identifiable and need to be extracted. Furthermore, we are focused on extracting tumor names that are associated with a therapeutic agent (i.e., has a drug-target) registered in the CTR, so that it enables us to obtain a clear view of the therapeutic and drug-target landscape for a given tumor in future studies. To address this need, we designed the CANTOS pipeline to extract tumor names from the rest of the conditions, and then annotate each unique tumor as pediatric or adult tumors (**Figure 2**). The tumor name identification step implemented by CANTOS is segregated into three phases.

In Phase 1, CANTOS extracted condition names (and consequently tumor names) linked with CTR registered therapeutic agents that potentially had an associated drug-target. It achieves this by subsetting the condition names in the conditions file using the types of drugs from the interventions file. The interventions file contains 786,898 records and includes the fields: "id", "nct_id", "intervention_type", "name", and "description". The "id" represents the record number, while the "nct_id" serves as the foreign key. The "name" and "description" fields detail the interventions, and "intervention_type" classifies them into eleven categories listed in **Supplementary Document SD2**. CANTOS extracts condition names (consequently tumors names) linked with therapeutic agents potentially associated with drug-targets by joining the intervention file with the conditions file via "nct_id" and then filtering for the intervention types: "Drug", "Biological", "Combination Product", and "Genetic". Filtering therapeutic agents by these intervention types increases the likelihood that each agent has a corresponding drug

target. Consequently, any tumor name extracted this way can potentially highlight the therapeutic and drug-target landscape associated with that tumor. After filtering, CANTOS extracted 50,410 unique condition names, which served as input for tumor identification in Phase 2 of the pipeline. The tumor extraction process is illustrated as Phase 1 in **Figure 3**.

In Phase 2, CANTOS identified tumor names from the Phase 1 extracted condition names using two independent protocols (**Figure 3**, “Phase 2”). The first protocol consisted of checking if each condition name contained a tumor key word listed in **Supplementary Document SD3**. If the condition name contained a tumor keyword, that condition was annotated as a potential tumor. The second protocol of CANTOS matched each condition name in the CTR to tumor names in the WHO 5th edition using a fuzzy string matching algorithm. If a condition name from the clinical trials exactly matched a term in the WHO 5th edition (**Supplementary Table ST3**), it was annotated as a tumor. If there were no exact matches, we performed a fuzzy match by calculating the generalized Levenshtein edit distance between the CTR condition name and each term in the WHO 5th edition. This was implemented using the `agrep` function in R ¹¹⁰, with a maximum distance set to 0.2. If a WHO 5th edition term fell within this distance, the clinical trial condition was flagged as a potential tumor. Once flagged, we manually validated these findings to confirm that the conditions marked as potential tumors were indeed tumors. A total of 13,230 unique (by string uniqueness) tumor names were extracted from the CTR at the end of Phase 2.

During Phase 3, CANTOS focused on determining which extracted tumor names were pediatric (see **Figure 3**, Phase 3) by applying a fuzzy string matching algorithm similar to that used in Phase 2. However, this time, the algorithm compared the tumor names against only the pediatric tumor names (**Supplementary Table ST11**), listed in the WHO 5th edition. After CANTOS flagged the tumors as pediatric or adult, we manually validated the results. For tumors confirmed as pediatric, we added citations from peer-reviewed literature, government websites, or articles from research institutions that specify the tumor as pediatric. If the tumor was found in the WHO system, we reported it as “Listed in WHO Ped Tumor” instead of providing a literature reference. From the 50,410 conditions, 13,230 were identified as tumors, with 6,324 classified as pediatric tumors. These annotations are recorded in the **Supplementary Table ST10**.

A cursory analysis of the identified tumor names revealed various discrepancies, including typographical errors, extraneous information, missing values, drug names instead of condition names, and multiple tumor names. Furthermore, many tumor names in the disease file did not adhere to standardized nomenclature from the WHO system or the NCI database. These unstandardized names hinder integration with other biomedical databases like IDG or OT. Additionally, they obstructed the manual annotation of tumors as pediatric or adult, leading to 144 tumors being labeled as “DA” (Do not Annotate) in the pediatric tumor designation field (“PedCanTumor”) in **Supplementary Table ST10**. **Supplementary Document SD4** highlights common discrepancies in tumor names. To address these issues, standardization is needed to link tumor names to external databases and gain insights into

associated drug targets and therapeutic agents. The standardization methods used by CANTOS are detailed in the following section.

Standardization of tumor names from CTR

We implemented various methods in CANTOS to standardize the tumor names in the CTR. These methods are based either on text-matching (edit-distances) or text-embedding. It also combined unsupervised clustering with text-matching and text-embeddings to evaluate if there is improvement in standardization. In total, CANTOS implemented 36 standardization methods, and their performance accuracies are discussed in the results section. The following subsections will detail the text-matching methods first, followed by the text-embedding techniques.

Text matching technique: closest match using edit distance

The aim of standardizing tumor names from the CTR to corresponding WHO system or NCIt database terms is to ensure each unstandardized term aligns accurately with its standardized equivalent while preserving the original meaning of the CTR tumor name. Edit distances measure the similarity between text strings by calculating the minimum number of edit operations (deletions, substitutions, insertions, etc.), q-grams, or heuristics needed to transform one string into another. The larger the edit distance between two strings, the further apart the strings are, thus, two strings with minimal edit distance could potentially convey the same meaning. CANTOS employs normalized Levenshtein distance, Jaro-Winkler distance, and cosine distance to compute edit distances. An example of using edit distances for string comparison is provided in **Supplementary Document SD5**, followed by brief descriptions of each method.

Normalized Levenshtein distance: Levenshtein distance measures the minimum number of single-character edits (insertions, deletions, substitutions) needed to transform one string into another. Levenshtein distance between two strings can be normalized by dividing it by the length of the longer string thus the distance would range from [0,1] and allow comparison of multiple strings on the same scale. We defined the normalized Levenshtein distance between two strings S1 and S2 as follows:

$$Distance_{Normalized\ Levenshtein} = \frac{Distance_{Levenshtein}}{\max(|S1|, |S2|)}$$

In the above equation, |S1| and |S2| represent the respective lengths of strings S1 and S2 between which we are computing the normalized Levenshtein distance. We calculate Levenshtein distance using the stringdist library in the R programming language^{111,112}. Following the calculation of the Levenshtein distance, we compute the normalizing factor (i.e. divide the Levenshtein by the longest string size) for distance between each pair of strings and normalize the Levenshtein distance.

Jarro-Winkler distance: The Jarro-Winkler distance is a normalized edit distance between two strings. It is a variant of the Jarro similarity measure which is defined as follows between two strings S1 and S2 respectively:

$$Sim_{jarro} = 0, \text{ if } m = 0$$

$$= \frac{1}{3} \left(\frac{m}{|S1|} + \frac{m}{|S2|} + \frac{m-t}{m} \right), \text{ otherwise}$$

Where $|S1|$ and $|S2|$ are lengths of the strings $S1$ and $S2$ respectively, m is the number of matching characters and t is the number of transpositions. It should be noted that when estimating 'm', each character in $S1$ compared to the characters in $S2$, and match is counted only when the characters are the same and if the characters are within a certain distance of each other typically defined as half the length of the longer string, rounded down, minus one, i.e. $\frac{\max(|S1|, |S2|)}{2} - 1$.

The Jarro-Winkler similarity measure builds on top of the Jarro similarity measure and introduces two more parameters for rewards and favorable scales the Jarro similarity score if the two strings share similar prefixes. The Jarro-Winkler similarity is defined as follows:

$$Sim_{jarro-winkler} = Sim_{jarro} + lp(1 - Sim_{jarro})$$

Where l is defined as the length of the common prefix at the start of the string (maximum of 4 characters), whereas p is a scaling factor that rewards the score for having common prefixes. Typically p is set to 0.1 and should not exceed 0.25 (or $\frac{1}{4}$ as the maximum length of prefix being considered is 4). With the above definition of Jarro-Winkler similarity, the Jarro-Winkler distance is defined as follows:

$$Distance_{jarro-winkler} = 1 - Sim_{jarro-winkler}$$

We calculate the Jarro-Winkler distance using the stringdist package in the R-programming language.

Cosine Distance: To define cosine distance, we first need to establish the concept of cosine similarity. For two non-zero vector vectors, cosine similarity is defined as the dot product of the two vectors divided by the product of their lengths. Cosine similarity ranges from $[-1,1]$, with -1 representing total opposition, 0 representing complete dissimilarity, and 1 representing full similarity between the vectors. Cosine similarity between two vectors A and B is defined as follows:

$$Sim_{cosine} = \cos(\theta) = \frac{A \cdot B}{|A| |B|}$$

However, to use cosine similarity in the context of strings, the vectors A and B represent the frequencies of unique words in strings $S1$ and $S2$. Since frequencies cannot be negative, the cosine similarity ranges between $[0,1]$. Thus, there is no need to normalize this metric, and cosine distance is defined simply as:

$$Distance_{cosine} = 1 - Sim_{cosine}$$

We calculate the cosine distance using the stringdist package in R. Based on the three edit distances, CANTOS computed the pairwise distances between each tumor name identified in the CTR and the standardized tumor terms with respect to the WHO system (5th edition and all editions) and NCIt database. For each CTR tumor name, CANTOS selects the nearest standardized terms under each edit distance. If more than one term qualified as the closest term, CANTOS reported them all by separating individual terms with a semicolon.

Text matching technique: edit distance combined with affinity propagation clustering

CANTOS implements another set of standardization methods based on edit distances and clustering techniques. These methods consist of a clustering step followed by a mapping step. CANTOS applied affinity propagation (AP) clustering to form the clusters, where the divergence matrix is computed by calculating the pairwise edit distance between the tumor names in CTR, WHO system, and NCIt database. CANTOS computes three divergence matrices using normalized Levenshtein distance, Jarro-Winkler distance, and cosine distance. Using each of these divergence matrices, CANTOS runs the AP clustering and forms three sets of clusters. We selected the AP algorithm for clustering as it automatically determines the number of clusters instead of requiring the number of clusters (unknown to us) to be a user-defined hyperparameter. AP is also not dependent on the initialization conditions and is deterministic¹¹³. AP works by recursively passing real-valued messages between each data point until they converge, and based on these converged values, the algorithm establishes the clusters and assigns each cluster an "exemplar data point" which serves as an ideal representative of that cluster¹¹⁴. Furthermore, AP clustering methods have shown success in clustering textual data^{115,116}. Once the clusters were computed using AP, CANTOS performed a cluster size analysis to check if any clusters were large and may contain members that should not belong together. This was done by determining the median cluster size, and clusters larger than the median cluster size were identified and designated as large clusters. On each of these large clusters, CANTOS performs nested AP clustering until their sizes drop below the previously determined median cluster size or if the AP clustering algorithm converges and no more new clusters can be performed.

CANTOS then detects outliers in each cluster using isolation forest and local outlier factors (LOF) algorithms. Outliers identified by either method are removed from their clusters and reassigned as single-member clusters. CANTOS implements isolation forest using the R isolation.forest package¹¹⁷, where we set the hyperparameter number of trees (ntrees) to 100 as recommended by Lie et al.2008¹¹⁸ in their original introduction of the isolation forest algorithm and the dims argument to 3, as suggested for numeric datasets in the package documentation¹¹⁷. The standardized outlier scores are calculated for each data point within a cluster, a score close to 1 indicates the data point is a likely outlier, while a score close to 0 indicates the data point is likely a member of the cluster. The CANTOS pipeline uses an outlier score of 0.5 as the threshold and any data point with an outlier score greater than 0.5 is deemed as an outlier. Similarly, CANTOS uses the lof function within the dbscan package¹¹⁹ in R to compute the LOF values to determine outliers in each cluster. The LOF value for a data point p is defined as the local reachability density p and the local reachability density of "minPts"-nearest neighbors of p ¹²⁰⁻¹²². The

hyperparameter “minPts” specifies the minimum number of nearest data points around p that need to be considered for calculating LOF value. An LOF value close to 1 indicates that the data point p is in a region with a relatively uniform density, whereas a $\text{LOF} > 1$ indicates the data point p has a lower density than its neighbors and is likely an outlier¹²². To compute the LOF values for each cluster member in each cluster, the CANTOS pipeline requires the users to define the minPts parameter. We set minPts to be integers ranging from 2 (clusters need to have more than one element to have an outlier) to $\text{Size}(\text{cluster})-1$. Iterating through each value of minPts, CANTOS then computes LOF values for each cluster element and then computes their median LOF value. If the median LOF value is above 1 for a data point p in a cluster, then it is designated as an outlier.

Upon completing the outlier analysis, CANTOS implements the mapping step, in which each cluster member is mapped to a standardized term. To achieve this, CANTOS iterates through each cluster and identifies the closest standardized term from the WHO system and NCI database based on the edit distance implemented in the pipeline. If most of the tumor names in a cluster are close to a specific standardized term, then all the tumor names are mapped to that standardized term. In case there is a tie, where two or more standardized terms are equally represented in a cluster, then each tumor name within that cluster is assigned to its closest (based on the edit distance used so far) matching standardized tumor name. The text-based matching pipeline is described in detail in **Supplementary Figure SF1** and the summarized version is shown in **Figure 4**. The following section will discuss the pipeline that standardizes the clinical trial tumors based on text-embedding.

Text embedding analysis: closest match in embedding space

The methods in the previous section employed edit distances to compare texts. These methods primarily focus on syntactical differences to quantify the differences between texts. In this section, the standardization methods deployed by CANTOS are based on text embeddings (or word embeddings), which can also be used for comparing texts. Text embeddings are low dimensional numeric vector representations of unstructured text data. Unlike edit distances, text embeddings focus on capturing the semantic and contextual meaning of the input text they encode; consequently, in the embedding vector space, texts with similar meanings should have embeddings close to each other and texts which differ in meaning should be further apart^{86,123–125}. Text-embeddings have been used in various applications such as developing search engines^{126,127}, text clustering¹²⁸ and classification¹²⁹, recommender systems¹³⁰, and anomaly detection¹³¹. Text-embeddings can be generated by natural language processing models such as Word2Vec¹²⁴, GloVE¹³², FastText^{133–135} or through large language models (LLM) such as BERT¹³⁶, GPT^{137–141}, ELMO^{86,142} or specialized transformer based models such as text-embedding-3-large (LTE-3)⁸⁴, text-embedding-ada-002 (ADA-002)⁸³ etc.

For this study, we generated text-embeddings from 26 different transformer based models and LLMs, many of which were open source and a few of them were proprietary models. These models varied in their architectures, training data, domain specialization and text-embedding dimensions. The open source models used for generating text-embeddings include e5-large^{47,48}, e5-large-v2^{47,49}, all-mpnet-base-v2⁵⁰, all-MiniLM-L12-v2⁵¹, gtr-t5-large^{52,53}, all-MiniLM-L6-v2⁵⁴, all-roberta-large-v1⁵⁵,

sapBERT^{56,57}, clinicalBERT^{58,59}, PubMedBERT^{60,61}, BioBERT⁶², SciBERT^{63,64}, ModernBERT^{65,66}, BioGPT^{67,68}, MedLlama2^{69,70}, MedLlama_13B⁷¹, LaBSE^{72,73}, Llama3.3_70B^{74,75}, Llama3.2_3B^{75,76}, Llama3.0^{75,77}, Phi-4^{78,79} nomic⁸⁰, and DeepSeek_8B^{81,82}. The proprietary models include OpenAI's: text-embedding-ada-002 (ADA-002)⁸³ and text-embedding-3-large (LTE-3)⁸⁴ and Cohere's embed-english-v2.0⁸⁵. The proprietary models of Open AI: ADA-002, LTE-3 and Cohere's embed-english-v2.0 are accessible through their API and there is a cost for using the model which is based on the amount of computational resources consumed. Furthermore, since the models are proprietary in nature, the architecture and weights are not disclosed or available for download to the public. Thus the users have very limited control over the model and fine tuning of these models are generally limited. In contrast, open-source models are not subjected to these limitations. They allow users to execute the models locally, customize them according to specific requirements, and operate without incurring usage costs. Further details on each of these text-embeddings and their models is available in **Table 5**. We used all of the above models to generate text-embeddings for tumor names in the CTR, WHO system, and NCIt database and provide it as input to CANTOS. For each text-embedding, CANTOS computes the pairwise Euclidean distance between all the tumor names contained in the CTR, WHO system and NCIt database. Following this step, for each tumor name in the CTR, CANTOS identifies and maps it to the nearest standardized term from the WHO 5th edition, WHO all editions and NCIt database.

Text embedding analysis: OpenAI models embeddings and clustering

We implemented a set of methods that integrate clustering with text-embedding, similar to the approach that combines AP clustering with edit distances discussed in the preceding section. The objective behind implementing these methods in CANTOS was to evaluate whether clustering could offer significant improvements in standardization accuracy compared to the closest match method used for tumor name standardization. We limited these methods to use Open AI's text-embedding models LTE-3 and ADA-002 as they achieved high and consistent standardization accuracy against both the WHO 5th and all editions. Furthermore, combining clustering with text-embeddings makes these methods computationally expensive, which prompted us to limit these methods to use LTE-3 and ADA-002 text-embeddings.

Similar to CANTOS's approach of using edit distances to compute divergence matrices for AP clustering, in this standardization approach, CANTOS first calculated the pairwise Euclidean distances in the embedding space (LTE-3 or ADA-002) among tumor names in the CTR, WHO system and NCIt database. However, this step is computationally expensive due to the high dimensionality of the embedding space: ADA-002 (1536 dimensions) and LTE-3 (3072 dimension). To address this issue, CANTOS applied principal component analysis (PCA) to each set of embeddings (ADA-002 and LTE-3) and based on the iteration of pipeline that depended on the edition of WHO (5th edition or all editions) system used in the pipeline, generated four sets of PCA-transformed embeddings. The dimensions of the four PCA-transformed embedding spaces are summarized in **Table 6**. For each case, CANTOS retained the minimum number of principal components necessary to explain 80% of the variance in the data.

With the four sets of PCA-transformed embeddings, CANTOS computed the divergence matrices for AP clustering using pairwise Euclidean distance. After clustering, CANTOS performs cluster size analysis by computing the z-scores for each cluster based on the number of cluster members. Using the default z-score of 2.5 as a threshold, CANTOS determined the maximum cluster size and designated any cluster with a z-score greater than 2.5 as a large cluster. For each large cluster, CANTOS performed nested AP clustering until their sizes were below or equal to the maximum number of cluster members or the AP clustering algorithm converged. The use of z-score for determining large clusters differs from the edit distance-based AP clustering method in the previous section, where the median cluster size was used as the threshold for determining the large clusters in CANTOS. The median cluster size for embedding-based AP clustering was lower than the maximum cluster size established by z-score-based clustering. This consequently led to more clusters with relatively homogeneous cluster members being labeled as large clusters and flagged for nested clustering. The maximum cluster size determined using the z-score was larger, and fewer such clusters were flagged as large, hence it was selected as the threshold for deciding large clusters in CANTOS. It should be noted that CTR has a diverse range of tumor names; thus, clustering these tumor names will produce clusters of nonuniform cluster size. Hence, estimating a maximum cluster size is not trivial when considering the variation in the entire dataset. With more curation and classification of the tumors within the CTR, one can get a more accurate estimate of a reasonable cluster size for each tumor type.

After clustering, CANTOS performed outlier detection as in the previous section using isolation forest and LOF analysis. The hyperparameters for isolation forest and LOF analysis were kept the same as they were for edit-distance-based AP clustering. Following the outlier detection step, CANTOS iterates through each cluster and determines the standardized term from the WHO system (5th edition and all editions) and the NCIt database that are closest to each cluster member by computing the Euclidean distance in the embedding space (not the PCA-transformed space). If a majority of the tumor names in a cluster are close to a specific standardized term, then all the tumor names are mapped to that standardized term. In case there is a tie, where two or more standardized terms are equally represented in a cluster, then each tumor name within that cluster is assigned to its closest matching standardized term. **Table 7** compares the number of AP clusters formed when we use text-embeddings and text-matching (edit distances).

In addition to AP clustering, CANTOS implemented K-means clustering on PCA transformed embeddings to standardize the tumor names in CTR. Unlike AP, K-means requires users to define the number of clusters, "K," as a hyperparameter¹⁴³. Without prior information on the types of tumors present in CTR, it is difficult to estimate "K" qualitatively and thus we used a computational method to decide on an optimal value for "K". To determine "K", we implemented a commonly used cluster performance metric known as the silhouette coefficient¹⁴⁴ within CANTOS. For a given value of "K", when K-means clustering is completed, the silhouette coefficient is computed for each data point. The silhouette coefficient has a range of [-1,1]. A silhouette coefficient of 1 signifies that the data point is

well-matched to other elements in its own cluster and poorly matched to members of neighboring clusters¹⁴⁵. A silhouette coefficient of 0 indicates that the data point is at the decision boundary of neighboring clusters and a score of -1 indicates that the data point is poorly matched with other cluster members and likely assigned to an incorrect cluster¹⁴⁵. Intuitively, a higher silhouette coefficient for a data point represents high cohesion of that data point with rest of the cluster members and high separation from members of neighboring clusters. CANTOS iterates through several values of “K” and performs K-means clustering and calculates the silhouette coefficient for each data point. For each cluster, CANTOS calculates the average silhouette coefficient which indicates the performance for that cluster. CANTOS then computed the average of each of the averaged silhouette coefficients per cluster, this produces a metric that can be used to evaluate the overall clustering performance of the K-means algorithm for a given value of “K”. We refer to this metric as the mean silhouette score which ranges from [-1,1] and a higher positive value indicates better clustering performance. Using this method, CANTOS evaluated the mean silhouette score for various values of “K” under both ADA-002 and LTE-3 embeddings for each iteration (WHO 5th edition and WHO all editions). In **Figure 5**, we plot the mean silhouette score for each of these cases.

After K-means clustering is completed, CANTOS follows the same steps for outlier detection (isolation forest and LOF analysis) and standardization (compute Euclidean distances with standardized term and identify the closest match) as it did for embedding based AP clustering, while maintaining the exact hyperparameter configurations. A detailed view of the text-embedding based standardization pipeline is displayed in **Supplementary Figure SF2**, and a brief summary is presented in **Figure 4**.

Standardization using majority-vote of high-accuracy standardization methods:

Based on the performance of the aforementioned standardization methods, we developed an approach that combines three high-accuracy standardization methods to improve overall standardization accuracy. This approach standardizes a given CTR tumor name by assigning it to the term most frequently predicted among the three methods (majority vote). Since eleven methods qualified as high-accuracy standardization techniques for both the WHO 5th edition and all editions combined, we can theoretically test for 165 ($^{11}C_3$) possible three-method combinations for either edition of the WHO system (See Tables 1 and 2). However, rather than testing all possible combinations, we prioritized selecting three methods with high disagreement (i.e., high diversity in their predictions) in the prediction. This strategy aims to maximize the standardization accuracy by focusing on selecting methods in a way such that when one method incorrectly standardizes a CTR tumor name, the other methods have a higher chance of correctly standardizing the same. To select the three methods, we first computed the pairwise mutual information among the eleven standardization methods.

To select the methods using this criteria, we create a matrix containing the pairwise mutual information of each high-accuracy method using their standardization results generated against the 1600 randomly sampled tumor names from CTR under both the WHO 5th edition and all editions. **Figures 6 and 7** display the pairwise mutual information heatmap for the WHO 5th edition and all editions respectively.

The corresponding pairwise mutual information values are provided in **Supplementary Tables ST12 and ST13**.

The lowest pairwise mutual information is between the methods all-MiniLM-L12-v2 +Euclidean Dist and LTE-3+K-means for both WHO 5th edition and all editions. Following the creation of the mutual information matrix, for each of the 165 combinations of standardization methods we calculated the mean pairwise mutual information. Since each combination contains three methods, the pairwise mutual information of each of these methods (${}^3C_2=3$) with respect are selected from the mutual information matrix and then averaged. The combination of standardization methods with the lowest mutual information (i.e., they share the least amount of information and are therefore more diverse) are selected for standardization using the majority vote. In the event of tie, we select the method with the highest standardization accuracy (depending on the WHO system edition) among the three methods that were selected to perform the majority vote.

In addition to the above mutual information-based majority voting standardization method, we also implemented a method which utilized all the eleven high-accuracy standardization methods (see **Tables 1 and 2**) for performing a majority vote prediction and selecting a standardized tumor name for a given CTR tumor name. In cases of ties, tumor names were assigned to the method with the highest standardization accuracy based on the respective edition of the WHO classification system. For the WHO 5th edition, ties were resolved using predictions from all-MiniLM-L12-v2 + Euclidean Dist, while for WHO all editions, the LTE-3 + Euclidean Dist method was used.

Conclusion

The CTR records various aspects of clinical trials, including conditions studied, and shares this information publicly in text file format. However, it lacks a mechanism for identifying tumor names, which are often unstandardized, making it challenging to integrate them with other biomedical databases for downstream analysis. In this paper, we present CANTOS, a computational pipeline that extracts tumor names from the condition file in CTR and classifies them as adult or pediatric tumors. It then standardizes the extracted tumor names with respect to the WHO system and NCIt database using 36 standardization methods based on text-matching and text-embedding. Our findings show that methods based on text-embedding have a varying range of standardization accuracy. Some of these methods outperformed text-matching methods, while others had similar or poorer performance compared to text-matching methods, indicating that not all text-embeddings capture the nuances and intricacies needed to map unstandardized tumor names from the CTR to their standardized form in the WHO system. The highest standardization accuracy was achieved by the text-embedding based methods LTE-3+Euclidean Dist when standardizing against WHO all editions and by all-MiniLM-L12-v2+Euclidean Dist when standardizing against WHO 5th edition. We standardized all 13,230 tumors in the CTR against the WHO system and NCIt database and reported the results.

Our findings suggest the need for using richer text-embeddings obtained from models trained and fine tuned on large diverse biomedical and oncology-related data and the WHO resource. Text-embedding from such a model will likely improve the identification and classification of non-standard tumor names, potentially enhancing the accuracy and robustness of the CANTOS pipeline.

Resource availability

Data and code availability

Users can download the data used in this paper from the CTR website (<https://clinicaltrials.gov/>)⁹, Clinical Trials API or from the Aggregate Analysis of ClinicalTrials.gov-Clinical Trials Transformative Initiative (AACT-CTTI) website (<https://aact.ctti-clinicaltrials.org/download>). We downloaded a copy of the database from the AACT-CTTI website on August 22, 2023 which is available as a zip file titled “20230822_export.zip” under the section titled “Monthly Archive of Static Copies”. The AACT-CTTI website is updated daily with content from ClinicalTrials.gov and provides a static database at the beginning of each month. This database includes information on all registered studies in the CTR, with details about clinical trials, such as experimental design, conditions, and interventions, available in separate pipe-delimited text files within the zip file. The process for downloading the dataset used in this study is outlined in **Supplementary Document SD6**. The code files for building and running CANTOS, along with the instructions for downloading the embeddings generated using Open AI are publicly available at our GitHub repository: <https://github.com/TaylorResearchLab/CANTOS>.

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Deanne Taylor (taylorlrm@chop.edu).

Materials availability

The WHO system and NCI database standardized tumor names generated by CANTOS are available in **Supplementary Tables ST6-ST9**. The code for implementation of the CANTOS pipeline is available in our public GitHub repository: <https://github.com/TaylorResearchLab/CANTOS>.

Acknowledgements

Dr. Taylor would like to acknowledge funding from the NIH Common Fund, through the Office of Strategic Coordination/Office of the NIH Director under award U2CHD109731. All other funding was provided by the Children’s Hospital of Philadelphia’s Research Institute. We thank Dr. Susan Furth and CHOP’s Department of Biomedical and Health Informatics for their ongoing support, and Dr. Sarah Tasian for her guidance on cancer nomenclature review.

Author contributions

AL and DT conceived and designed the study. AL wrote the code. AL,SS, BS, TM performed the analyses. AL wrote the manuscript. DT, EM, and KB made critical revisions to the manuscript. All authors interpreted the content, made revisions to the manuscript, and had final approval of the completed version. DT oversaw the project. DT acquired funding for the project.

Declaration of interests

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT 4.0 and Grammarly to improve clarity and conciseness of the written text. After using these services, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

1. Bray, F., Laversanne, M., Sung, H., Ferlay, J., Siegel, R.L., Soerjomataram, I., and Jemal, A. (2024). Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J. Clin.* *74*, 229–263.
2. Siegel, R.L., Giaquinto, A.N., and Jemal, A. (2024). Cancer statistics, 2024. *CA Cancer J. Clin.* *74*, 12–49.
3. Matt, G.Y., Sioson, E., Shelton, K., Wang, J., Lu, C., Zaldivar Peraza, A., Gangwani, K., Paul, R., Reilly, C., Acić, A., et al. (2024). St. Jude Survivorship Portal: Sharing and Analyzing Large Clinical and Genomic Datasets from Pediatric Cancer Survivors. *Cancer Discov.* *14*, 1403–1417.
4. Aristizabal, P., Winestone, L.E., Umaretiya, P., and Bona, K. (2021). Disparities in Pediatric Oncology: The 21st Century Opportunity to Improve Outcomes for Children and Adolescents With Cancer. *Am Soc Clin Oncol Educ Book* *41*, e315–e326.
5. Hunger Stephen P., and Mullighan Charles G. Acute Lymphoblastic Leukemia in Children. *N. Engl. J. Med.* *373*, 1541–1552.
6. Laetsch, T.W., DuBois, S.G., Bender, J.G., Macy, M.E., and Moreno, L. (2021). Opportunities and Challenges in Drug Development for Pediatric Cancers. *Cancer Discov.* *11*, 545–559.
7. Renfro, L.A., Ji, L., Piao, J., Onar-Thomas, A., Kairalla, J.A., and Alonzo, T.A. (2019). Trial Design Challenges and Approaches for Precision Oncology in Rare Tumors: Experiences of the Children’s Oncology Group. *JCO Precis Oncol* *3*. <https://doi.org/10.1200/PO.19.00060>.
8. Rivers, Z., Hyde, B., Ronski, K., Stearns, D., Toll, S., Ritt, K., Cooney, M., Nimeiri, H., Federman, N., and Kaneva, K. (2023). Exploring Barriers to Pediatric Cancer Clinical Trials: The Role of a Networked, Just-in-Time Study Program. *Clin. Ther.* *45*, 1148–1150.

9. National Institutes of Health Clinical Trials Registry ClinicalTrials.gov.
<https://clinicaltrials.gov/>.
10. Zarin, D.A., Tse, T., Williams, R.J., Califf, R.M., and Ide, N.C. (2011). The ClinicalTrials.gov Results Database — Update and Key Issues. *N. Engl. J. Med.* 364, 852–860.
11. WHO Classification of Tumours Editorial Board ed. (2019). Digestive System Tumours 5th ed. (International Agency for Research on Cancer).
12. WHO Classification of Tumours Editorial Board ed. (2019). Breast Tumours 5th ed. (International Agency for Research on Cancer).
13. WHO Classification of Tumours Editorial Board ed. (2020). Soft Tissue and Bone Tumours 5th ed. (International Agency for Research on Cancer).
14. WHO Classification of Tumours Editorial Board ed. (2020). Female Genital Tumours 5th ed. (International Agency for Research on Cancer).
15. WHO Classification of Tumours Editorial Board ed. (2021). Thoracic Tumours 5th ed. (International Agency for Research on Cancer).
16. WHO Classification of Tumours Editorial Board ed. (2021). Central Nervous System Tumours 5th ed. (International Agency for Research on Cancer).
17. WHO Classification of Tumours Editorial Board ed. (2022). Urinary and Male Genital Tumours 5th ed. (International Agency for Research on Cancer).
18. WHO Classification of Tumours Editorial Board ed. (2023). Paediatric tumours: Who classification of tumours 5th ed. (International Agency for Research on Cancer).
19. WHO Classification of Tumours Editorial Board ed. (2024). Head and Neck Tumours 5th ed. (International Agency for Research on Cancer).
20. WHO Classification of Tumours Editorial Board ed. (2024). Haematolymphoid tumours: Part B 5th ed. (International Agency for Research on Cancer).
21. Elder DE, Massi D, Scolyer RA, Willemze R ed. (2018). WHO Classification of Skin Tumours 4th ed. (International Agency for Research on Cancer).
22. Grossniklaus, H.E., Eberhart, C.G., and Kivelä, T.T. eds. (2018). WHO Classification of Tumours of the Eye 4th ed. (International Agency for Research on Cancer).
23. Lloyd, R.V., Osamura, R.Y., Klöppel, G., and Rosai, J. eds. (2017). WHO Classification of Tumours of Endocrine Organs 4th ed. (International Agency for Research on Cancer).
24. El-Naggar AK, Chan JKC, Grandis JR, Takata T, Slootweg PJ ed. (2017). WHO Classification of Head and Neck Tumours 4th ed. (International Agency for Research on Cancer).

25. Swerdlow SH, Campo E, Harris NL, Jaffe ES, Pileri SA, Stein H, Thiele J ed. (2017). WHO classification of Tumours of Haematopoietic and Lymphoid Tissues Revised 4th Edition 4th ed. (International Agency for Research on Cancer).
26. Moch H, Humphrey PA, Ulbright TM, Reuter VE ed. (2016). WHO Classification of Tumours of the Urinary System and Male Genital Organs 4th ed. (International Agency for Research on Cancer).
27. Louis DN, Ohgaki H, Wiestler OD, Cavenee WK ed. (2016). WHO Classification of Tumours of the Central Nervous System (Revised 4th Edition) Revised 4th Edition. (International Agency for Research on Cancer).
28. Travis WD, Brambilla E, Burke AP, Marx A, Nicholson AG ed. (2015). WHO Classification of Tumours of the Lung, Plura, Thymus and Heart 4th ed. (International Agency for Research on Cancer).
29. Kurman RJ, Carcangiu ML, Herrington CS, Young RH ed. (2014). WHO Classification of Tumours of Female Reproductive Organs 4th ed. (International Agency for Research on Cancer).
30. Fletcher CDM, Bridge JA, Hogendoorn PCW, Mertens F ed. (2013). WHO Classification of Tumours of Soft tissue and Bone 4th ed. (International Agency for Research on Cancer).
31. van de Vijver MJ, L.S.E.I.S.S.T.P. ed. (2012). WHO Classification of Tumours of the Breast 4th ed. (International Agency for Research on Cancer).
32. Bosman FT, Carneiro F, Hruban RH, Theise ND ed. (2010). WHO Classification of Tumours of the Digestive System 4th ed. (International Agency for Research on Cancer).
33. Swerdlow SH, Campo E, Harris NL, Jaffe ES, Pileri SA, Stein H, Thiele J, Vardiman JW ed. (2008). WHO Classification of Tumours of Haematopoietic and Lymphoid Tissues 4th ed. (International Agency for Research on Cancer).
34. Louis DN, Ohgaki H, Wiestler OD, Cavenee WK ed. (2006). WHO Classification of Tumours of the Central Nervous System 4th ed. (International Agency for Research on Cancer).
35. Barnes L, Eveson JW, Reichart P, Sidransky D ed. (2005). Pathology and Genetics of Head and Neck Tumours 3rd ed. (International Agency for Research on Cancer).
36. LeBoit PE, Burg G, Weedon D, Sarasin A ed. (2005). Pathology and Genetics of Skin Tumours 3rd ed. (International Agency for Research on Cancer).
37. Travis WD, Brambilla E, Müller-Hermelink HK, Harris CC ed. (2004). Pathology and genetics of tumours of the lung, plura, thymus and heart 3rd ed. (International Agency for Research on Cancer).
38. DeLellis RA, Lloyd RV, Heitz PU, Eng C ed. (2004). Pathology and Genetics of Tumours of the Endocrine Organs 3rd ed. (International Agency for Research on Cancer).

39. Eble JN, Sauter G, Epstein J, Sesterhenn I ed. (2004). Pathology and Genetics of Tumours of the Urinary System and Male Genital Organs 3rd ed. (International Agency for Research on Cancer).
40. Tavassoli FA, D.P. ed. (2003). Pathology and Genetics of Tumours of the Breast and Female Genital Organs 3rd ed. (International Agency for Research on Cancer).
41. Pathology and Genetics of Tumours of Soft Tissue and Bone (2002). 3rd ed. (International Agency for Research on Cancer).
42. Jaffe ES, Harris NL, Stein H, Vardiman JW ed. (2001). Pathology and Genetics of Tumours of the Haematopoietic and Lymphoid Tissues 3rd ed. (International Agency for Research on Cancer).
43. Hamilton, A.L.A., Sr ed. (2000). Pathology and Genetics of Tumours of the Digestive System 3rd ed. (International Agency for Research on Cancer).
44. Kleihues P, C.W.K. ed. (2000). Pathology and Genetics of Tumours of the Nervous System 3rd ed. (International Agency for Research on Cancer).
45. Ochoa, D., Hercules, A., Carmona, M., Suveges, D., Gonzalez-Uriarte, A., Malangone, C., Miranda, A., Fumis, L., Carvalho-Silva, D., Spitzer, M., et al. (2021). Open Targets Platform: supporting systematic drug-target identification and prioritisation. *Nucleic Acids Res.* 49, D1302–D1310.
46. US National Institutes of Health (2014). Illuminating the Druggable Genome. Illuminating the Druggable Genome. <https://druggablegenome.net/>.
47. Wang, L., Yang, N., Huang, X., Jiao, B., Yang, L., Jiang, D., Majumder, R., and Wei, F. (2022). Text Embeddings by Weakly-Supervised Contrastive Pre-training. *arXiv [cs.CL]*.
48. Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, Furu Wei (2022). E5-Large. Hugging Face. <https://huggingface.co/intfloat/e5-large>.
49. Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, Furu Wei (2023). E5-large-v2. Hugging Face. <https://huggingface.co/intfloat/e5-large-v2>.
50. all-mpnet-base-v2 Hugging Face. <https://huggingface.co/sentence-transformers/all-mpnet-base-v2>.
51. all-MiniLM-L12-v2 Hugging Face. <https://huggingface.co/sentence-transformers/all-MiniLM-L12-v2>.
52. Ni, J., Qu, C., Lu, J., Dai, Z., Ábrego, G.H., Ma, J., Zhao, V.Y., Luan, Y., Hall, K.B., Chang, M.-W., et al. (2021). Large Dual Encoders Are Generalizable Retrievers. *arXiv [cs.IR]*.

53. Jianmo Ni, Chen Qu, Jing Lu, Zhuyun Dai, Gustavo Hernández Ábrego, Ji Ma, Vincent Y. Zhao, Yi Luan, Keith B. Hall, Ming-Wei Chang, Yinfei Yang (2021). gtr-t5-large. Hugging Face. <https://huggingface.co/sentence-transformers/gtr-t5-large>.
54. all-MiniLM-L6-v2 Hugging Face.
<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>.
55. all-roberta-large-v1 Hugging Face.
<https://huggingface.co/sentence-transformers/all-roberta-large-v1>.
56. Fangyu Liu, Ehsan Shareghi, Zaiqiao Meng, Marco Basaldella, Nigel Collier (2021). SapBERT. Hugging Face.
<https://huggingface.co/cambridgeltl/SapBERT-from-PubMedBERT-fulltext>.
57. Liu, F., Shareghi, E., Meng, Z., Basaldella, M., and Collier, N. (2020). Self-alignment pretraining for biomedical entity representations. *arXiv [cs.CL]*.
58. Wang, G., Liu, X., Ying, Z., Yang, G., Chen, Z., Liu, Z., Zhang, M., Yan, H., Lu, Y., Gao, Y., et al. (2023). Optimized glycemic control of type 2 diabetes with reinforcement learning: a proof-of-concept trial. *Nat. Med.* 29, 2633–2642.
59. Guangyu Wang, Xiaohong Liu, Zhen Ying, Guoxing Yang, Zhiwei Chen, Zhiwen Liu, Min Zhang, Hongmei Yan, Yuxing Lu, Yuanxu Gao, Kanmin Xue, Xiaoying Li, Ying Chen (2023). ClinicalBERT. Hugging Face. <https://huggingface.co/medicalai/ClinicalBERT>.
60. Gu, Y., Tinn, R., Cheng, H., Lucas, M., Usuyama, N., Liu, X., Naumann, T., Gao, J., and Poon, H. (2022). Domain-specific language model pretraining for biomedical natural language processing. *ACM Trans. Comput. Healthc.* 3, 1–23.
61. Yu Gu , Robert Tinn , Hao Cheng , Michael Lucas , Naoto Usuyama , Xiaodong Liu , Tristan Naumann and Jianfeng Gao , Hoifung Poon (2020). BiomedNLP-BiomedBERT-base-uncased-abstract. Hugging Face.
<https://huggingface.co/microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract>.
62. Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C.H., and Kang, J. (2020). BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics* 36, 1234–1240.
63. Beltagy, I., Lo, K., and Cohan, A. (2019). SciBERT: A pretrained language model for scientific text. *arXiv [cs.CL]*.
64. Cohan, I.B.K.L. (2019). scibert_scivocab_uncased. Hugging Face.
https://huggingface.co/allenai/scibert_scivocab_uncased.
65. Warner, B., Chaffin, A., Clavié, B., Weller, O., Hallström, O., Taghadouini, S., Gallagher, A., Biswas, R., Ladhak, F., Aarsen, T., et al. (2024). Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. *arXiv [cs.CL]*. <https://doi.org/10.48550/ARXIV.2412.13663>.

66. Benjamin Warner and Antoine Chaffin and Benjamin Clavié and Orion Weller and Oskar Hallström and Said Taghadouini and Alexis Gallagher and Raja Biswas and Faisal Ladhak and Tom Aarsen and Nathan Cooper and Griffin Adams and Jeremy Howard and Iacopo Poli (2024). ModernBERT-large. Hugging Face.
<https://huggingface.co/answerdotai/ModernBERT-large>.
67. Luo, R., Sun, L., Xia, Y., Qin, T., Zhang, S., Poon, H., and Liu, T.-Y. (2022). BioGPT: generative pre-trained transformer for biomedical text generation and mining. *Brief. Bioinform.* 23, bbac409.
68. Renqian Luo, Liai Sun, Yingce Xia, Tao Qin, Sheng Zhang, Hoifung Poon, Tie-Yan Liu (2022). BioGPT. Hugging Face. <https://huggingface.co/microsoft/biogpt>.
69. Raval, S. MedLlama2: Fine-tuned Llama 2 model to answer medical questions based on an open source medical dataset. https://huggingface.co/llSourcecell/medllama2_7b.
70. MedLlama 2: Fine-tuned Llama 2 model to answer medical questions based on an open source medical dataset Ollama. <https://ollama.com/library/medllama2>.
71. Wu, C. (2023). MedLLaMA13B. https://huggingface.co/chaoyi-wu/MedLLaMA_13B.
72. Feng, F., Yang, Y., Cer, D., Arivazhagan, N., and Wang, W. (2020). Language-agnostic BERT sentence embedding. *arXiv [cs.CL]*.
73. Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Arivazhagan, Wei Wang (2020). LaBSE. Hugging Face. <https://huggingface.co/sentence-transformers/LaBSE>.
74. Meta, A.I. (2024). Llama 3.3 70B: A multilingual instruction-tuned large language model. <https://ollama.com/library/llama3.3>.
75. Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., et al. (2023). LLaMA: Open and efficient foundation language models. *arXiv [cs.CL]*. <https://doi.org/10.48550/ARXIV.2302.13971>.
76. Meta, A.I. (2024). Llama 3.2: Revolutionizing edge AI and vision with open models. <https://ollama.com/library/llama3.2>.
77. Meta Llama 3: The most capable openly available LLM to date
<https://ollama.com/library/llama3>.
78. Microsoft Research (2024). Phi-4 Technical Report (Microsoft Corporation).
79. Phi-4 is a 14B parameter, state-of-the-art open model from Microsoft Ollama.
<https://ollama.com/library/phi4>.
80. Nussbaum, Z., Morris, J.X., Duderstadt, B., and Mulyar, A. (2024). Nomic embed: Training a reproducible long context text embedder. *arXiv [cs.CL]*.
<https://doi.org/10.48550/ARXIV.2402.01613>.

81. Deepseek (2025). Depseek-r1:8b. Ollama. <https://ollama.com/library/deepseek-r1:8b>.
82. DeepSeek-AI, Guo, D., Yang, D., Zhang, H., Song, J., Zhang, R., Xu, R., Zhu, Q., Ma, S., Wang, P., et al. (2025). DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. arXiv [cs.CL]. <https://doi.org/10.48550/ARXIV.2501.12948>.
83. text-embedding-ada-002 (2023). OpenAI. <https://openai.com/index/new-and-improved-embedding-model/>.
84. New embedding models and API updates <https://openai.com/index/new-embedding-models-and-api-updates/>.
85. Cohere embed-english-v2.0 (2023). Cohere. <https://docs.cohere.com/v2/docs/cohere-embed>.
86. Khattak, F.K., Jeblee, S., Pou-Prom, C., Abdalla, M., Meaney, C., and Rudzicz, F. (2019). A survey of word embeddings for clinical text. *J. Biomed. Inform.* 100S, 100057.
87. Ghoshal, P., and Rao, X. (2019). Normalization of shorthand forms in French text messages using word embedding and machine translation. In *Parallel Corpora for Contrastive and Translation Studies* (John Benjamins Publishing Company), pp. 281–298.
88. Popovski, G., Paudel, B., Eftimov, T., and Seljak, B.K. (2019). Exploring a standardized language for describing foods using embedding techniques. In *2019 IEEE International Conference on Big Data (Big Data) (IEEE)*, pp. 5172–5176.
89. Zhang, Y., Ma, X., and Song, G. (2018). Chinese medical concept normalization by using text and comorbidity network embedding. In *2018 IEEE International Conference on Data Mining (ICDM) (IEEE)*, pp. 777–786.
90. Kasmaee, A.S., Khodadad, M., Saloot, M.A., Sherck, N., Dokas, S., Mahyar, H., and Samiee, S. (2024). ChemTEB: Chemical Text Embedding Benchmark, an overview of embedding models performance & efficiency on a specific domain. arXiv [cs.CL].
91. Vasilevsky, N.A., Matentzoglou, N.A., Toro, S., Flack, J.E., IV, Hegde, H., Unni, D.R., Alyea, G.F., Amberger, J.S., Babb, L., Balhoff, J.P., et al. (2022). Mondo: Unifying diseases for the world, by the world. <https://doi.org/10.1101/2022.04.13.22273750>.
92. Baron, J.A., Johnson, C.S.-B., Schor, M.A., Olley, D., Nickel, L., Felix, V., Munro, J.B., Bello, S.M., Bearer, C., Lichenstein, R., et al. (2024). The DO-KB Knowledgebase: a 20-year journey developing the disease open science ecosystem. *Nucleic Acids Res.* 52, D1305–D1314.
93. Schriml, L.M., Lichenstein, R., Bisordi, K., Bearer, C., Baron, J.A., and Greene, C. (2023). Modeling the enigma of complex disease etiology. *J. Transl. Med.* 21, 148.
94. Baron, J.A., and Schriml, L.M. (2023). Assessing resource use: a case study with the Human Disease Ontology. *Database (Oxford)* 2023. <https://doi.org/10.1093/database/baad007>.

95. Schriml, L.M., Munro, J.B., Schor, M., Olley, D., McCracken, C., Felix, V., Baron, J.A., Jackson, R., Bello, S.M., Bearer, C., et al. (2022). The Human Disease Ontology 2022 update. *Nucleic Acids Res.* *50*, D1255–D1261.
96. Schriml, L.M., Mitraka, E., Munro, J., Tauber, B., Schor, M., Nickle, L., Felix, V., Jeng, L., Bearer, C., Lichenstein, R., et al. (2019). Human Disease Ontology 2018 update: classification, content and workflow expansion. *Nucleic Acids Res.* *47*, D955–D962.
97. Bello, S.M., Shimoyama, M., Mitraka, E., Laulederkind, S.J.F., Smith, C.L., Eppig, J.T., and Schriml, L.M. (2018). Disease Ontology: improving and unifying disease annotations across species. *Dis. Model. Mech.* *11*. <https://doi.org/10.1242/dmm.032839>.
98. Schriml, L.M., and Mitraka, E. (2015). The Disease Ontology: fostering interoperability between biological and clinical human disease-related data. *Mamm. Genome* *26*, 584–589.
99. Wu, T.-J., Schriml, L.M., Chen, Q.-R., Colbert, M., Crichton, D.J., Finney, R., Hu, Y., Kibbe, W.A., Kincaid, H., Meerzaman, D., et al. (2015). Generating a focused view of disease ontology cancer terms for pan-cancer data integration and analysis. *Database (Oxford)* *2015*, bav032.
100. Kibbe, W.A., Arze, C., Felix, V., Mitraka, E., Bolton, E., Fu, G., Mungall, C.J., Binder, J.X., Malone, J., Vasant, D., et al. (2015). Disease Ontology 2015 update: an expanded and updated database of human diseases for linking biomedical knowledge through disease data. *Nucleic Acids Res.* *43*, D1071–D1078.
101. Schriml, L.M., Arze, C., Nadendla, S., Chang, Y.-W.W., Mazaitis, M., Felix, V., Feng, G., and Kibbe, W.A. (2012). Disease Ontology: a backbone for disease semantic integration. *Nucleic Acids Res.* *40*, D940–D946.
102. Osborne, J.D., Flatow, J., Holko, M., Lin, S.M., Kibbe, W.A., Zhu, L.J., Danila, M.I., Feng, G., and Chisholm, R.L. (2009). Annotating the human genome with Disease Ontology. *BMC Genomics* *10 Suppl 1*, S6.
103. Köhler, S., Gargano, M., Matentzoglou, N., Carmody, L.C., Lewis-Smith, D., Vasilevsky, N.A., Danis, D., Balagura, G., Baynam, G., Brower, A.M., et al. (2021). The Human Phenotype Ontology in 2021. *Nucleic Acids Res.* *49*, D1207–D1217.
104. World Health Organization (2016). International Statistical Classification of Diseases and Related Health Problems. 10th Revision (ICD-10). 5th ed. <https://icd.who.int/browse10/>.
105. SNOMED CT: Systematized Nomenclature of Medicine Clinical Terms SNOMED International. SNOMED International. <https://www.snomed.org/>.
106. Stanford CRFM <https://crfm.stanford.edu/2022/12/15/biomedlm.html>.
107. Canonica, G.W., Baena-Cagnani, C.E., Bousquet, J., Bousquet, P.J., Lockey, R.F., Malling, H.-J., Passalacqua, G., Potter, P., and Valovirta, E. (2007). Recommendations for standardization of clinical trials with Allergen Specific Immunotherapy for respiratory

allergy. A statement of a World Allergy Organization (WAO) taskforce. *Allergy* 62, 317–324.

108. Katz, M.H.G., Marsh, R., Herman, J.M., Shi, Q., Collison, E., Venook, A.P., Kindler, H.L., Alberts, S.R., Philip, P., Lowy, A.M., et al. (2013). Borderline resectable pancreatic cancer: need for standardization and methods for optimal clinical trial design. *Ann. Surg. Oncol.* 20, 2787–2795.
109. Dickersin, K., and Mayo-Wilson, E. (2018). Standards for design and measurement would make clinical research reproducible and usable. *Proc. Natl. Acad. Sci. U. S. A.* 115, 2590–2594.
110. Snášel, V., Kepřt, A., Abraham, A., and Hassanien, A.E. (2009). Approximate String Matching by Fuzzy Automata. In *Man-Machine Interactions* (Springer Berlin Heidelberg), pp. 281–290.
111. van der Loo, M. (2013). Stringdist: Approximate string matching, fuzzy text search, and string distance functions. (The R Foundation). <https://cran.r-project.org/package=stringdist>
<https://cran.r-project.org/package=stringdist>.
112. R Core Team (2024) *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>.
113. Frey, B.J., and Dueck, D. (2007). Clustering by passing messages between data points. *Science* 315, 972–976.
114. Kitahara, Y.F.G.I. Fast Algorithm for Affinity Propagation. In *Twenty-Second International Joint Conference on Artificial Intelligence*, T. Walsh, ed. (IJCAI/AAAI), pp. 2238–2243.
115. Shi, X.H., Guan, R.C., Wang, L.P., Pei, Z.L., and Liang, Y.C. (2009). An incremental affinity propagation algorithm and its applications for text clustering. In *2009 International Joint Conference on Neural Networks* (IEEE). <https://doi.org/10.1109/ijcnn.2009.5178973>.
116. Shailendra Kumar Shrivastava, J.L.Rana, R.C.Jain (2013). Text document clustering based on phrase similarity using affinity propagation. *International Journal of Computer Applications* 61. <https://doi.org/10.5120/10032-5077>.
117. Cortes, D. (2019). isotree: Isolation-Based Outlier Detection. (The R Foundation). <https://doi.org/10.32614/cran.package.isotree> <https://doi.org/10.32614/cran.package.isotree>.
118. Liu, F.T., Ting, K.M., and Zhou, Z.-H. (2008). Isolation Forest. In *2008 Eighth IEEE International Conference on Data Mining* (IEEE), pp. 413–422.
119. Hahsler, M., and Piekenbrock, M. (2015). Dbscan: Density-based spatial clustering of applications with noise (DBSCAN) and related algorithms. (The R Foundation). <https://doi.org/10.32614/cran.package.dbscan> <https://doi.org/10.32614/cran.package.dbscan>.
120. Alghushairy, O., Alsini, R., Soule, T., and Ma, X. (2020). A review of Local Outlier Factor algorithms for outlier detection in big data streams. *Big Data Cogn. Comput.* 5, 1.

121. Ding, H., Ding, K., Zhang, J., Wang, Y., Gao, L., Li, Y., Chen, F., Shao, Z., and Lai, W. (2018). Local outlier factor-based fault detection and evaluation of photovoltaic system. *Solar Energy* 164, 139–148.
122. Xu, H., Zhang, L., Li, P., and Zhu, F. (2022). Outlier detection algorithm based on k-nearest neighbors-local outlier factor. *J. Algorithm. Comput. Technol.* 16. <https://doi.org/10.1177/17483026221078111>.
123. Morris, J., Kuleshov, V., Shmatikov, V., and Rush, A. (2023). Text embeddings reveal (almost) as much as text. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (Association for Computational Linguistics)*. <https://doi.org/10.18653/v1/2023.emnlp-main.765>.
124. Mikolov, T. (2013). Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*.
125. Incitti, F., Urli, F., and Snidaro, L. (2023). Beyond word embeddings: A survey. *Inf. Fusion* 89, 418–436.
126. Gökçe, O., Prada, J., Nikolov, N.I., Gu, N., and Hahnloser, R.H.R. (2020). Embedding-based scientific literature discovery in a text editor application. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics: System Demonstrations (Association for Computational Linguistics)*. <https://doi.org/10.18653/v1/2020.acl-demos.36>.
127. Mai, G., Janowicz, K., and Yan, B. (2018). Combining text embedding and knowledge graph embedding techniques for academic search engines. In *Joint proceedings of the 4th Workshop on Semantic Deep Learning (SemDeep-4) and NLIWoD4: Natural Language Interfaces for the Web of Data (NLIWOD-4) and 9th Question Answering over Linked Data challenge (QALD-9) co-located with 17th International Semantic Web Conference (ISWC 2018) CEUR Workshop Proceedings.*, Key-Sun Choi, Luis Espinosa Anke, Thierry Declerck, Dagmar Gromann, Jin-Dong Kim, Axel-Cyrille Ngonga Ngomo, Muhammad Saleem, Ricardo Usbeck, ed. (CEUR), pp. 77–88.
128. Mehta, V., Bawa, S., and Singh, J. (2021). WEClustering: word embeddings based text clustering technique for large datasets. *Complex Intell Systems* 7, 3211–3224.
129. Stein, R.A., Jaques, P.A., and Valiati, J.F. (2019). An analysis of hierarchical text classification using word embeddings. *Inf. Sci.* 471, 216–232.
130. Musto, C., Semeraro, G., de Gemmis, M., and Lops, P. (2016). Learning word embeddings from Wikipedia for content-based recommender systems. In *Lecture Notes in Computer Science Lecture notes in computer science*. (Springer International Publishing), pp. 729–734.
131. Pande, A., and Ahuja, V. (2017). WEAC: Word embeddings for anomaly classification from event logs. In *2017 IEEE International Conference on Big Data (Big Data) (IEEE)*, pp. 1095–1100.
132. Jeffrey Pennington , Richard Socher , Christopher D. Manning (2014). GloVe: Global Vectors

for Word Representation. In Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), A. M. B. P. Daelemans, ed. (Association for Computational Linguistics), pp. 1532–1543.

133. Bojanowski, P., Grave, E., Joulin, A., and Mikolov, T. (2016). Enriching word vectors with subword information. arXiv [cs.CL].
134. Joulin, A., Grave, E., Bojanowski, P., and Mikolov, T. (2016). Bag of tricks for efficient text classification. arXiv [cs.CL].
135. Joulin, A., Grave, E., Bojanowski, P., Douze, M., Jégou, H., and Mikolov, T. (2016). FastText.zip: Compressing text classification models. arXiv [cs.CL].
136. Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2018). BERT: Pre-training of deep bidirectional Transformers for language understanding. arXiv [cs.CL].
<https://doi.org/10.48550/ARXIV.1810.04805>.
137. OpenAI, Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F.L., Almeida, D., Altenschmidt, J., Altman, S., et al. (2023). GPT-4 Technical Report. arXiv [cs.CL].
<https://doi.org/10.48550/ARXIV.2303.08774>.
138. Brown, T.B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al. (2020). Language Models are Few-Shot Learners. arXiv [cs.CL]. <https://doi.org/10.48550/ARXIV.2005.14165>.
139. Yenduri, G., M, R., G, C.S., Y, S., Srivastava, G., Maddikunta, P.K.R., G, D.R., Jhaveri, R.H., B, P., Wang, W., et al. (2023). Generative Pre-trained transformer: A comprehensive review on enabling technologies, potential applications, emerging challenges, and future directions. arXiv [cs.CL].
140. Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever (2019). Language Models are Unsupervised Multitask Learners (OpenAI).
141. Alec Radford, Karthik Narasimhan, Tim Salimans, Ilya Sutskever (2018). Improving Language Understanding by Generative Pre-Training (OpenAI).
142. Matthew E. Peters, Mark Neumann, Mohit Iyyer, Matt Gardner, Christopher Clark, Kenton Lee, Luke Zettlemoyer (2018). Deep contextualized word representations. In Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers), M. W. H. J. Stent, ed. (Association for Computational Linguistics).
143. Wu, J. (2012). Cluster Analysis and K-means Clustering: An Introduction. In Advances in K-means Clustering: A Data Mining Thinking, J. Wu, ed. (Springer Berlin Heidelberg), pp. 1–16.
144. Shahapure, K.R., and Nicholas, C. (2020). Cluster Quality Analysis Using Silhouette Score. In 2020 IEEE 7th International Conference on Data Science and Advanced Analytics (DSAA)

(IEEE), pp. 747–748.

145. Shutaywi, M., and Kachouie, N.N. (2021). Silhouette Analysis for Performance Evaluation in Machine Learning with Applications to Clustering. *Entropy* 23.
<https://doi.org/10.3390/e23060759>.

TABLES

Table 1: CANTOS Accuracies for Standardization Methods when WHO system all editions are used.

Ranking	Basis	Methods	Accuracy WHO system all editions	Model Performance Classification
1	Embedding	LTE-3+Euclidean Dist	69.1%	High Accuracy
2	Embedding	e5-large+Euclidean Dist	68.2%	High Accuracy
3	Embedding	LTE-3+AP	67.5%	High Accuracy
4	Embedding	all-mpnet-base-v2+Euclidean Dist	67.4%	High Accuracy
5	Embedding	ADA-002+Euclidean Dist	66.7%	High Accuracy
6	Embedding	all-MiniLM-L6-v2+Euclidean Dist	66%	High Accuracy
7	Embedding	all-MiniLM-L12-v2+Euclidean Dist	66%	High Accuracy
8	Embedding	ADA-002+AP	65.2%	High Accuracy
9	Embedding	LTE-3+K-means	64.9%	High Accuracy
10	Embedding	ADA-002+K-means	64%	High Accuracy
11	Embedding	nomic+Euclidean Dist	62.2%	High Accuracy
12	Embedding	all-roberta-large-v1+Euclidean Dist	59.1%	Intermediate Accuracy
13	Embedding	gtr-t5-large+Euclidean Dist	58%	Intermediate Accuracy
14	Embedding	e5-large-v2+Euclidean Dist	56.3%	Intermediate Accuracy
15	Embedding	embed-english-v2.0+Euclidean Dist	53.6%	Intermediate Accuracy
16	Embedding	SapBERT+Euclidean Dist	47.9%	Low Accuracy
17	Embedding	ClinicalBERT+Euclidean Dist	40.1%	Low Accuracy
18	Embedding	BioGPT+Euclidean Dist	39.6%	Low Accuracy
19	Embedding	LaBSE+Euclidean Dist	35.2%	Low Accuracy
20	Text-Matching	Levenshtein	32.6%	Low Accuracy

21	Embedding	MedLlama_13B+Euclidean Dist	29.8%	Low Accuracy
22	Text-Matching	Levenshtein+AP	29.3%	Low Accuracy
23	Embedding	Llama3.3_70B+Euclidean Dist	27.2%	Low Accuracy
24	Text-Matching	Jarro Winkler	25.5%	Low Accuracy
25	Text-Matching	Jarro Winkler+AP	24.4%	Low Accuracy
26	Text-Matching	Cosine	23.9%	Low Accuracy
27	Embedding	DeepSeek_8B+Euclidean Dist	23.4%	Low Accuracy
28	Text-Matching	Cosine+AP	22.7%	Low Accuracy
29	Embedding	PubMedBERT+Euclidean Dist	22.5%	Low Accuracy
30	Embedding	SciBERT+Euclidean Dist	21.4%	Low Accuracy
31	Embedding	BioBERT+Euclidean Dist	21.3%	Low Accuracy
32	Embedding	Llama3.2_3B+Euclidean Dist	21.1%	Low Accuracy
33	Embedding	Llama3.0+Euclidean Dist	20.3%	Low Accuracy
34	Embedding	ModernBERT+Euclidean Dist	19.3%	Low Accuracy
35	Embedding	Phi-4+Euclidean Dist	17.2%	Low Accuracy
36	Embedding	MedLlama2+Euclidean Dist	15.4%	Low Accuracy

Table 2: CANTOS Accuracies for Standardization Methods when the WHO system 5th edition was used.

Ranking	Basis	Methods	Accuracy WHO system 5th edition	Model Performance Classification
1	Embedding	all-MiniLM-L12-v2+Euclidean Dist	67.8%	High Accuracy
2	Embedding	e5-large+Euclidean Dist	67.4%	High Accuracy
3	Embedding	all-mpnet-base-v2+Euclidean Dist	67.1%	High Accuracy
4	Embedding	all-MiniLM-L6-v2+Euclidean Dist	66.5%	High Accuracy
5	Embedding	LTE-3+Euclidean Dist	65.4%	High Accuracy
6	Embedding	LTE-3+AP	64.4%	High Accuracy
7	Embedding	LTE-3+K-means	63.4%	High Accuracy
8	Embedding	ADA-002+Euclidean Dist	62.6%	High Accuracy
9	Embedding	ADA-002+AP	62.5%	High Accuracy
10	Embedding	nomic+Euclidean Dist	60.8%	High Accuracy
11	Embedding	ADA-002+K-means	60.2%	High Accuracy
12	Embedding	e5-large-v2+Euclidean Dist	58.7%	Intermediate Accuracy
13	Embedding	all-roberta-large-v1+Euclidean Dist	58.6%	Intermediate Accuracy
14	Embedding	gtr-t5-large+Euclidean Dist	57%	Intermediate Accuracy
15	Embedding	embed-english-v2.0+Euclidean Dist	53.5%	Intermediate Accuracy
16	Embedding	SapBERT+Euclidean Dist	51.7%	Intermediate Accuracy
17	Embedding	ClinicalBERT+Euclidean Dist	40%	Low Accuracy
18	Embedding	BioGPT+Euclidean Dist	39.7%	Low Accuracy
19	Embedding	LaBSE+Euclidean Dist	36.1%	Low Accuracy
20	Embedding	MedLlama_13B+Euclidean Dist	31.9%	Low Accuracy
21	Text-Matching	Levenshtein	30.3%	Low Accuracy

22	Text-Matching	Levenshtein+AP	28.4%	Low Accuracy
23	Embedding	Llama3.3_70B+Euclidean Dist	27.4%	Low Accuracy
24	Text-Matching	Jarro Winkler	23.2%	Low Accuracy
25	Text-Matching	Jarro Winkler+AP	23%	Low Accuracy
26	Text-Matching	Cosine+AP	21.7%	Low Accuracy
27	Text-Matching	Cosine	21.6%	Low Accuracy
28	Embedding	DeepSeek_8B+Euclidean Dist	21.4%	Low Accuracy
29	Embedding	PubMedBERT+Euclidean Dist	21%	Low Accuracy
30	Embedding	Llama3.2_3B+Euclidean Dist	20.5%	Low Accuracy
31	Embedding	BioBERT+Euclidean Dist	20%	Low Accuracy
32	Embedding	LLama3.0+Euclidean Dist	19.8%	Low Accuracy
33	Embedding	SciBERT+Euclidean Dist	19.3%	Low Accuracy
34	Embedding	ModernBERT+Euclidean Dist	18.2%	Low Accuracy
36	Embedding	Phi-4+Euclidean Dist	15.4%	Low Accuracy
36	Embedding	MedLlama2+Euclidean Dist	13.7%	Low Accuracy

Table 3: Summary statistics for closest Euclidean distances in Figure 1. This table shows the summary statistics for closest Euclidean distances in **Figure 1** between the unstandardized CTR tumor names and standardized terms from the WHO 5th and all editions as visualized in box plots in **Figure 1** for the method LTE-3 + Euclidean Dist. 1,044 CTR tumor names had a ground truth in WHO system 5th edition and 1,118 CTR tumor names had a ground truth in WHO system all editions.

Summary Statistic	WHO system 5th edition		WHO system All editions	
	Correctly Standardized	Incorrectly Standardized	Correctly Standardized	Incorrectly Standardized
Minimum	0.000	0.339	0.000	0.394
1st Quartile	0.586	0.764	0.564	0.733
Median	0.696	0.841	0.687	0.825
Mean	0.666	0.826	0.647	0.811
3rd Quartile	0.804	0.908	0.802	0.897
Maximum	1.010	1.090	1.080	1.060

Table 4: Summary statistics for closest Euclidean distances in Figure 1. This table shows the summary statistics for closest Euclidean distances in **Figure 1** between the unstandardized CTR tumor names and standardized terms from the WHO 5th and all editions as visualized in box plots in **Figure 1** for the method all_MiniLM_L12_v2+ Euclidean Dist. 1,044 CTR tumor names had a ground truth in WHO system 5th edition and 1,118 CTR tumor names had a ground truth in WHO system all editions.

	WHO system 5th edition		WHO system All editions	
Summary Statistic	Correctly Standardized	Incorrectly Standardized	Correctly Standardized	Incorrectly Standardized
Minimum	0.000	0.163	0.000	0.163
1 st Quartile	0.442	0.628	0.418	0.603
Median	0.574	0.736	0.548	0.719
Mean	0.553	0.729	0.521	0.708
3 rd Quartile	0.700	0.828	0.669	0.817
Maximum	1.010	1.140	0.972	1.190

Table 5: Text embedding models properties.

	Embedding	Type	Accessibility	Embedding Dimensions	Model Parameters
1	Open AI: text-embedding-3-large (LTE-3)	Proprietary	https://platform.openai.com/docs/guides/embeddings	3072	Based on GPT-3 which has 175 billion parameters
2	Open AI: text-embedding-ada-002 (ADA-002)	Proprietary	https://platform.openai.com/docs/guides/embeddings	1536	Undisclosed
3	Cohere: embed-english-v2.0	Proprietary	https://docs.cohere.com/v2/docs/cohere-embed	4096	Undisclosed
4	LLama3.3 (Llama3.3_70B)	Open Source	https://ollama.com/library/llama3.3	8192	70 Billion
5	Llama3.2 (Llama3.2_3B)	Open Source	https://ollama.com/library/llama3.2	3072	3 Billion
6	LLama3.0	Open Source	https://ollama.com/library/llama3	4096	8 Billion
7	MedLLama2	Open Source	https://ollama.com/library/medllama2	4096	7 Billion
8	MedLLama_13B	Open Source	https://huggingface.co/chaoyi-wu/MedLLaMA_13B	5120	13 Billion
9	BioBERT	Open Source	https://github.com/dmis-lab/biobert-pytorch/blob/master/README.md	768	110 Million
10	PubMedBERT-Abstract (PubMedBERT)	Open Source	https://huggingface.co/microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract	768	110 Million
11	ModernBERT-Large (ModernBERT)	Open Source	https://huggingface.co/answerdotai/ModernBERT-large	1024	395 Million
12	Phi-4	Open Source	https://ollama.com/library/phi4	5120	14 Billion
13	e5-large	Open Source	https://huggingface.co/intfloat/e5-large	1024	335 Million

14	e5-large-v2	Open Source	https://huggingface.co/intfloat/e5-large-v2	1024	335 Million
15	all-mpnet-base-v2	Open Source	https://huggingface.co/sentence-transformers/all-mpnet-base-v2	768	109 Million
16	gtr-t5-large	Open Source	https://huggingface.co/sentence-transformers/gtr-t5-large	768	335 Million
17	all-MiniLM-L12-v2	Open Source	https://huggingface.co/sentence-transformers/all-MiniLM-L12-v2	384	33.4 Million
18	all-MiniLM-L6-v2	Open Source	https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2	384	22.7 Million
19	all-roberta-large-v1	Open Source	https://huggingface.co/sentence-transformers/all-roberta-large-v1	1024	355 Million
20	SapBERT	Open Source	https://huggingface.co/cambridgeltl/SapBERT-from-PubMedBERT-fulltext	768	109 Million
21	clinicalBERT	Open Source	https://huggingface.co/medicalai/ClinicalBERT	768	110 Million
22	LaBSE	Open Source	https://huggingface.co/sentence-transformers/LaBSE	768	Approx 470 Million
23	BioGPT	Open Source	https://huggingface.co/microsoft/biogpt	1024	355 Million
24	DeepSeek_8B	Open Source	https://ollama.com/library/deepseek-r1:8b	4096	8 Billion
25	sciBERT	Open Source	https://huggingface.co/allenai/scibert_scivocab_uncased	768	Undisclosed
26	nomic-embed-text	Open Source	https://ollama.com/library/nomic-embed-text	768	137 Million

Table 6: PCA dimensions used for each embedding space based on WHO system editions in CANTOS.

Tumor Terms	Dimensions for PCA+ADA-002	Dimensions for PCA+LTE-3
CTR + NCIT + WHO system 5th edition	136	178
CTR + NCIT + WHO system all editions	141	185

Table 7: Number of clusters from AP clustering under embedding and text-matching based methods.

Basis	Affinity Propagation Clustering Divergence Metric	Number of Clusters for CTR Terms, NCIt Terms, WHO system all editions	Number of Clusters for CTR Terms, NCIt Terms, WHO system 5th edition
Text Match	Cosine	1040	967
Text Match	Levenshtein	2020	1808
Text Match	Jarro Winkler	1965	1785
Embedding	ADA-002 + Euclidean Dist	3790	3456
Embedding	LTE-3 + Euclidean Dist	3894	3427

FIGURES

Figure 1: Distribution of closest Euclidean distances in the LTE-3 embedding space by the standardization method LTE-3+Euclidean Dist. This figure illustrates the distribution of the closest Euclidean distances in the LTE-3 embedding space, used by the standardization method LTE-3+Euclidean Dist, to map tumor names from the CTR to the nearest term from the WHO system. This distance is the minimum Euclidean distance between a given tumor name in the CTR and all the standardized terms in the WHO system within the LTE-3 embedding space. For a given tumor name from the CTR, the method LTE-3 + Euclidean Dist uses this distance to identify the nearest standardized term from the WHO system and then maps it to the respective tumor name in the CTR. The box plot in red shows the instances when the LTE-3 + Euclidean Dist method correctly standardized the CTR terms, whereas the blue box plot shows the instances when the method incorrectly standardized the CTR terms. **a.** displays the distribution of the closest Euclidean distances in LTE-3 embedding space when the CTR terms are standardized against the WHO system 5th edition, and **b.** displays the distribution of the closest Euclidean distances in LTE-3 embedding space when the CTR terms are standardized against WHO system all editions.

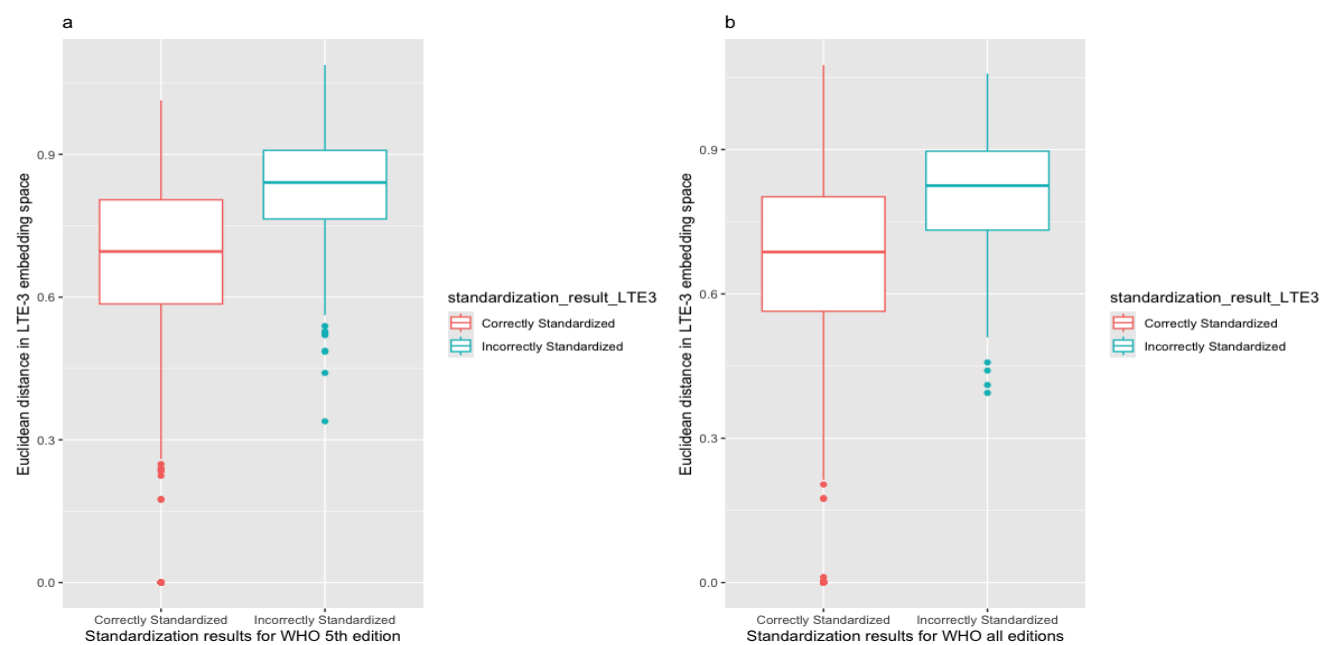


Figure 2: Distribution of closest Euclidean distances in the all-MiniLM-L12-v2 embedding space by the standardization method all-MiniLM-L12-v2+Euclidean Dist. This figure illustrates the distribution of the closest Euclidean distances in the all-MiniLM-L12-v2 embedding space, used by the standardization method all-MiniLM-L12-v2+Euclidean Dist, to map tumor names from the CTR to the nearest term from the WHO system. This distance is the minimum Euclidean distance between a given tumor name in the CTR and all the standardized terms in the WHO system within the all-MiniLM-L12-v2 embedding space. For a given tumor name from the CTR, the method all-MiniLM-L12-v2+Euclidean Dist uses this distance to identify the nearest standardized term from the WHO system and then maps it to the respective tumor name in the CTR. The box plot in red shows the instances when the all-MiniLM-L12-v2 + Euclidean Dist method correctly standardized the CTR terms, whereas the blue box plot shows the instances when the method incorrectly standardized the CTR terms. **a.** displays the distribution of the closest Euclidean distances in all-MiniLM-L12-v2 embedding space when the CTR terms are standardized against the WHO system 5th edition, and **b.** displays the distribution of the closest Euclidean distances in all-MiniLM-L12-v2 embedding space when the CTR terms are standardized against WHO system all editions.

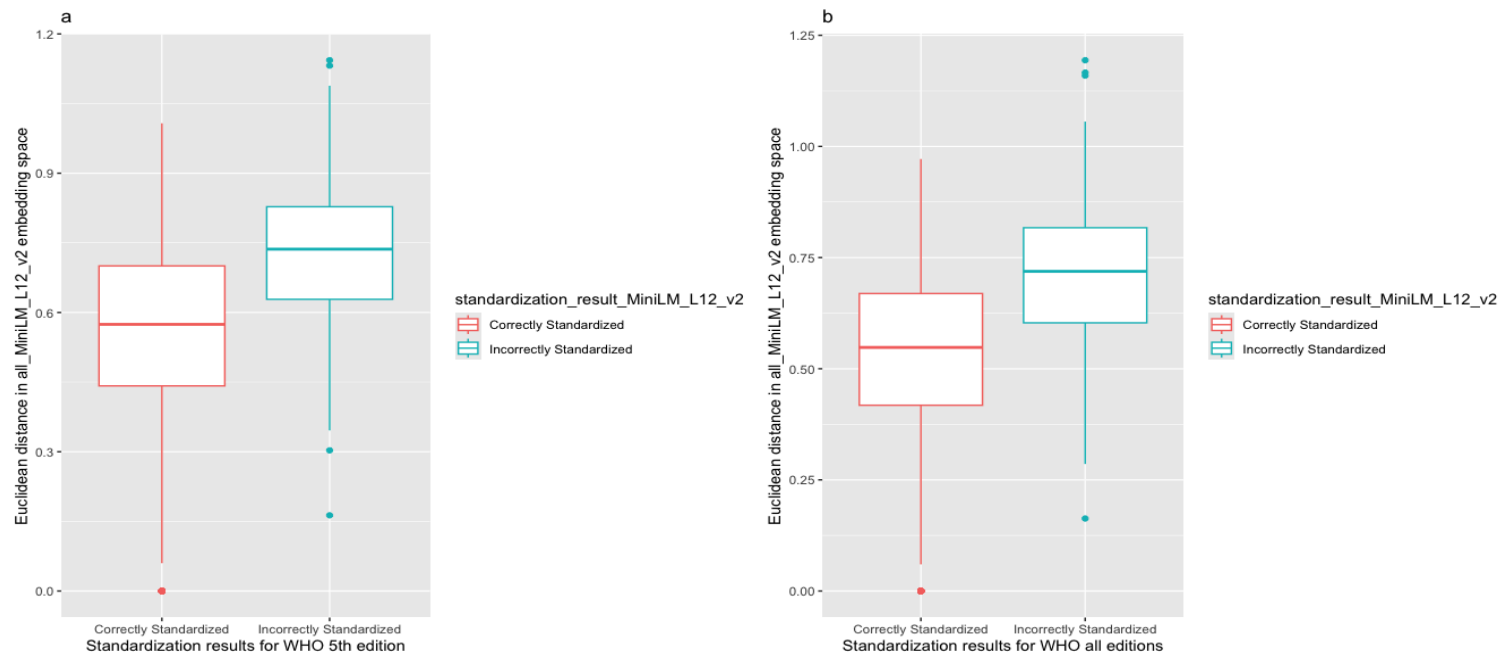


Figure 3: Tumor Annotation Process in the CANTOS Pipeline: The annotation workflow within the CANTOS pipeline, specifically for categorizing tumor names extracted from the NIH Clinical Trials Registry. The process involves filtering extracted tumor names, matching them to the WHO system, and categorizing them as adult or pediatric tumors. The conditions flagged as tumors were manually reviewed and citations were added for the pediatric tumor types.

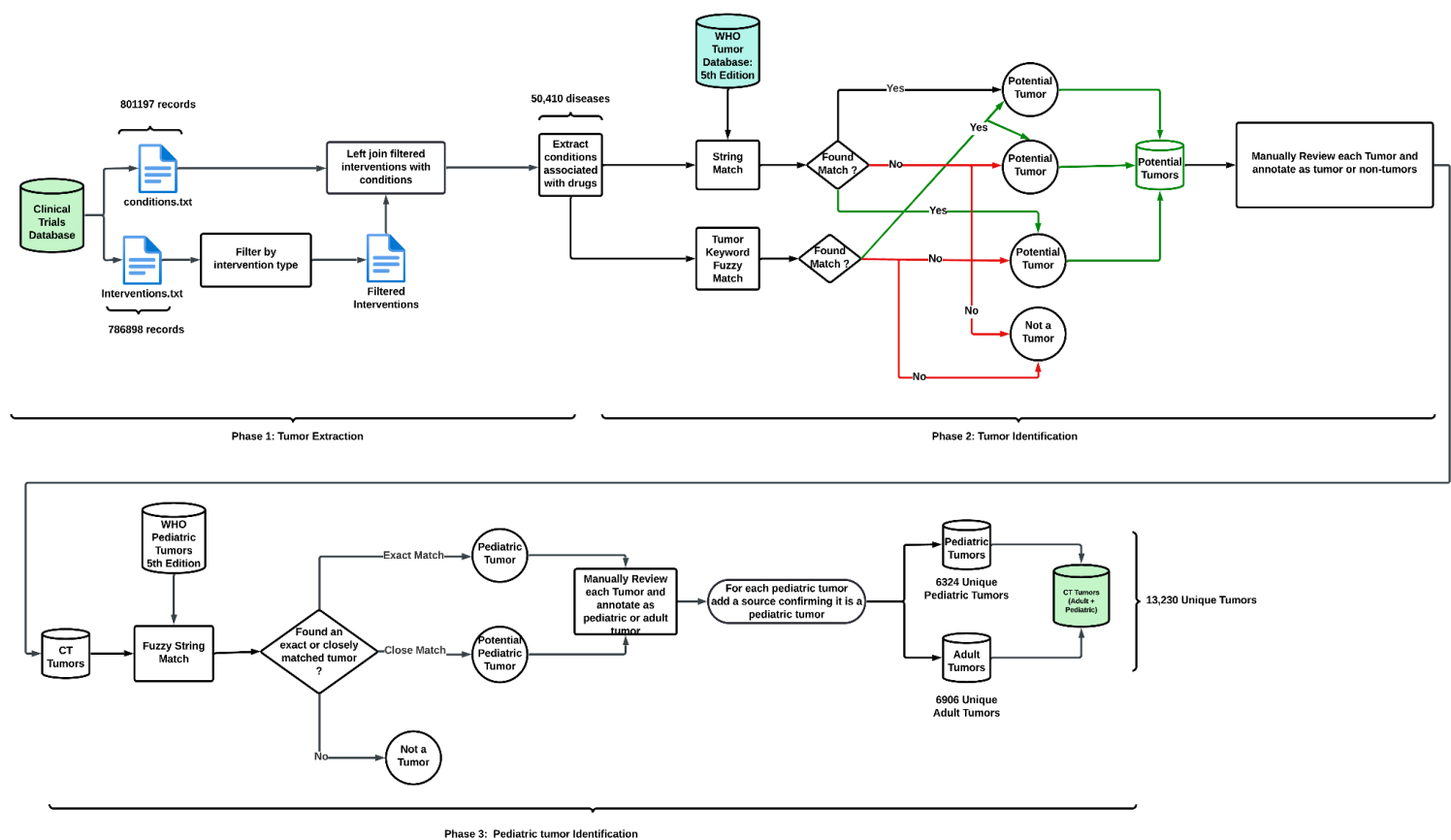


Figure 4: Text Matching and Embedding Pipeline for Tumor Standardization in CANTOS. Designed to standardize tumor names extracted from the NIH Clinical Trials Registry. The pipeline uses both edit-distance-based text matching and embedding-based methods to standardize the registered tumor names with entries in the WHO system and NCIt database.

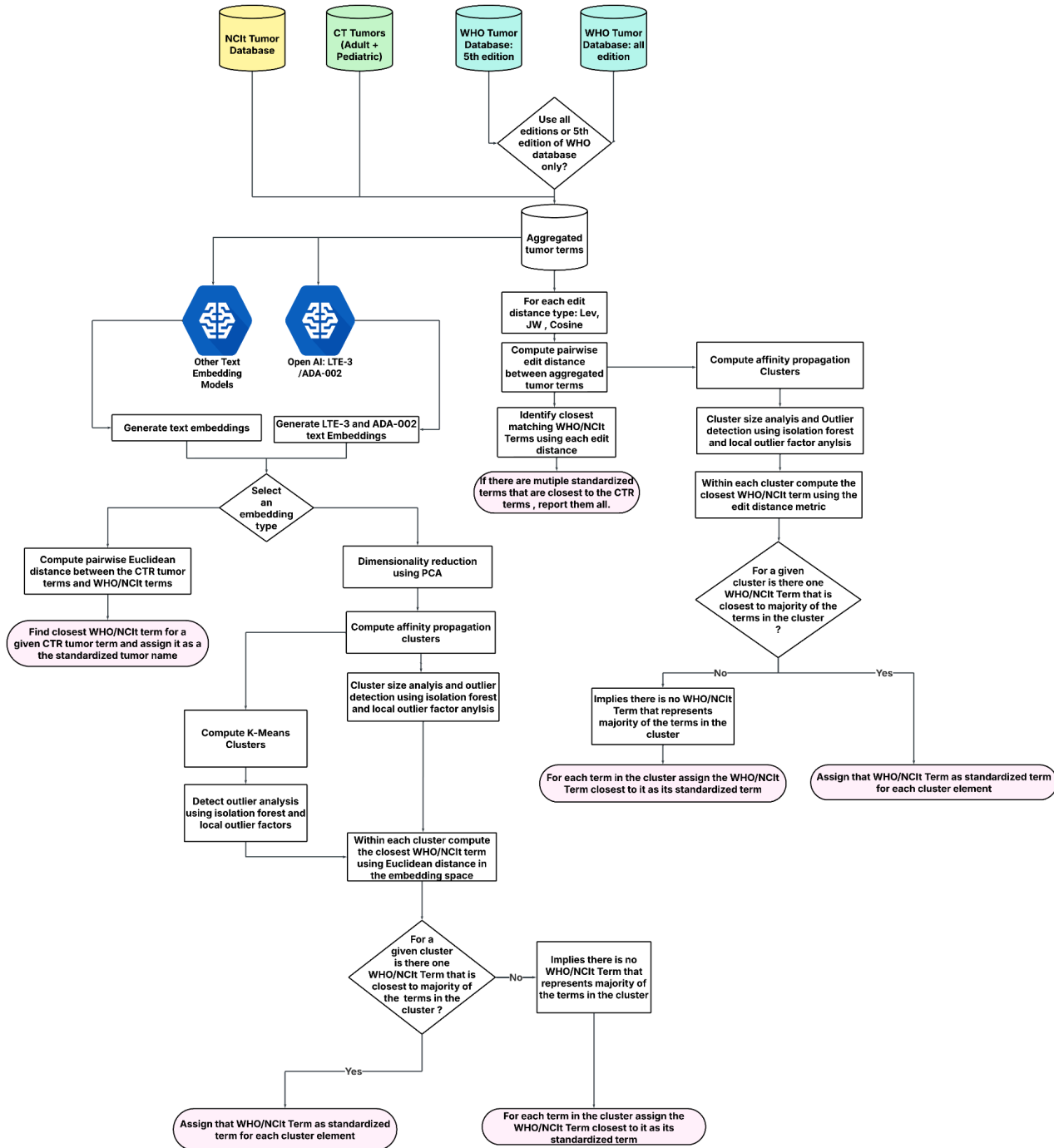
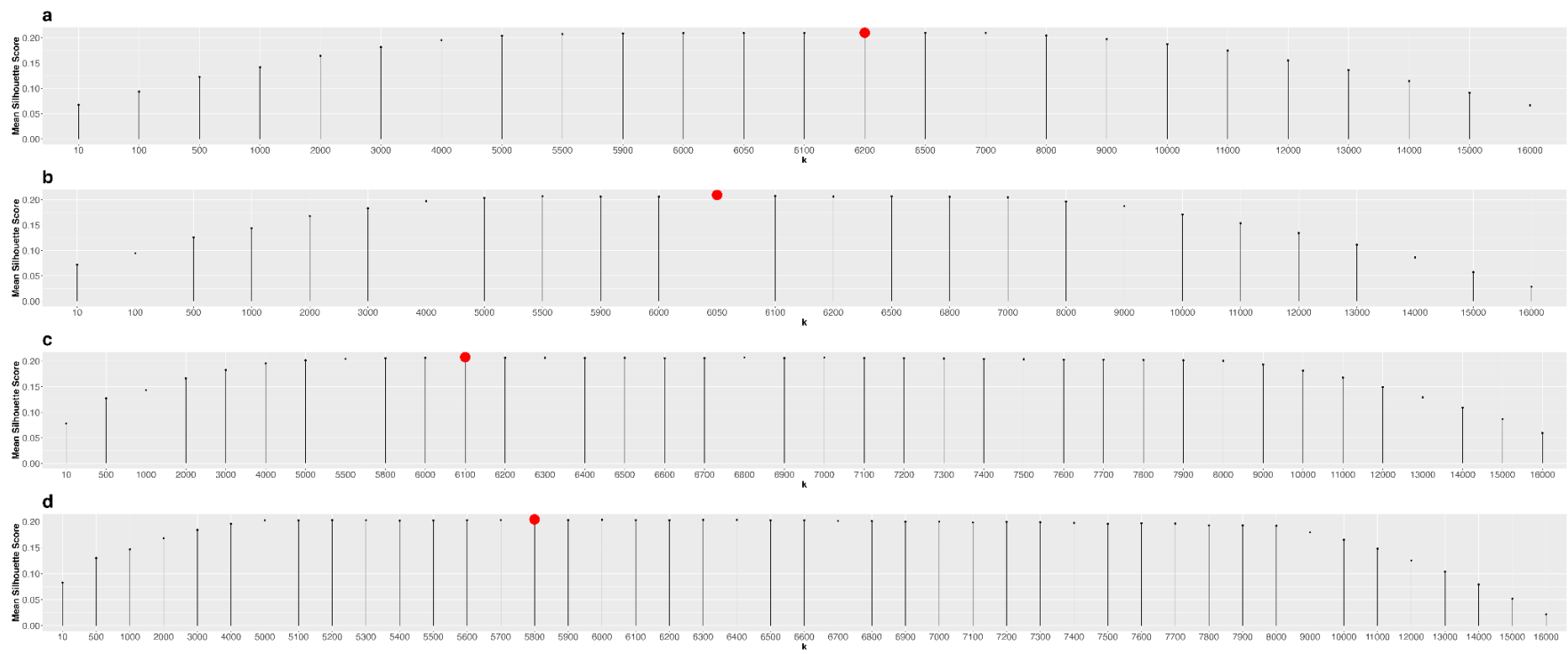


Figure 5: Average Silhouette Score vs Number of clusters (K) used in K-means. **a.** ADA-002 when WHO system all editions terms are used. **b.** ADA-002 when WHO system 5th edition terms are used **c.** LTE-3 when WHO system all editions terms are used **d.** LTE-3 when WHO system 5th editions terms are used.



Supplemental Information

Figures

Supplementary Figure SF1: SF1_CANTOS_Standardization_Workflow_Edit_Distances.pdf. This file contains a detailed view of the workflow employed by CANTOS to standardize the CTR tumor names using methods based on text-matching (edit distances).

Supplementary Figure SF2: SF2_CANTOS_Standardization_Workflow_Embedding.pdf. This file contains a detailed view of the workflow employed by CANTOS to standardize CTR tumor names using methods based on text-embedding.

Tables

Supplementary Table ST1: ST1_WHO_Tumor_all_edition.xlsx. This file contains the standardized tumor names from the WHO Tumor Classification System 3rd, 4th, and 5th editions.

Supplementary Table ST2: ST2_NCIT_Neoplasm_Core_terms.csv. This file contains the standardized tumor names from the NCIt database.

Supplementary Table ST3: ST3_who_cancer_key_words_general_5th_edition.xlsx. This file contains the standardized tumor names from the WHO Tumor Classification System 5th edition only.

Supplementary Table ST4: ST4_tumor_manually_validated_all.csv. This file contains the ground truth annotation with respect to all editions (3rd, 4th, and 5th) of WHO Tumor Classification System for the 1600 tumor names randomly sampled from the CTR. Furthermore, this file also contains the standardization results obtained from the 36 standardization methods implemented by CANTOS.

Supplementary Table ST5: ST5_tumor_manually_validated_5th.csv. This file contains the ground truth annotation with respect to the 5th edition of WHO Tumor Classification System for the 1600 tumor names randomly sampled from the CTR. Furthermore, this file also contains the standardization results obtained from the 36 standardization methods implemented by CANTOS.

Supplementary Table ST6: ST6_WHO_Results_all_ose.csv. This file contains the standardized tumor names for each CTR tumor identified by CANTOS. The standardization was done with respect to all editions (3rd, 4th, and 5th) of WHO Tumor Classification System. This result was obtained when all editions (3rd, 4th, and 5th) of WHO Tumor Classification System was in the CANTOS pipeline.

Supplementary Table ST7: ST7_WHO_Results_5thed_ose.csv. This file contains the standardized tumor names for each CTR tumor identified by CANTOS. The standardization was done with respect to the 5th edition of WHO Tumor Classification System. This result was obtained when the 5th edition of WHO Tumor Classification System was in the CANTOS pipeline.

Supplementary Table ST8: ST8_NCIT_Results_all_ose.csv. This file contains the standardized tumor names for each CTR tumor identified by CANTOS. The standardization was done with respect to the NCI database. This result was obtained when all editions (3rd, 4th, and 5th) of WHO Tumor Classification System was in the CANTOS pipeline.

Supplementary Table ST9: ST9_NCIT_Results_5thed_ose.csv. This file contains the standardized tumor names for each CTR tumor identified by CANTOS. The standardization was done with respect to the NCI database. This result was obtained when the 5th edition of WHO Tumor Classification System was in the CANTOS pipeline.

Supplementary Table ST10: ST10_tumor_annotated_adult_ped.csv. This file contains the list of tumors identified using CANTOS from the conditions file in CTR. For each condition name, the file informs the user if it is a tumor and if it is also a pediatric tumor. For each pediatric tumor, the file also provides a literature citation confirming that it is a pediatric tumor.

Supplementary Table ST11: ST11_who_cancer_key_words_paediatric_5th_edition.xlsx. This file contains the standardized pediatric tumor names from the WHO Tumor Classification System 5th edition only.

Supplementary Table ST12: ST12_MI_5th_results.csv. This file contains the pairwise mutual information between the eleven high-accuracy methods evaluated against the WHO Tumor Classification System 5th edition.

Supplementary Table ST13: ST13_MI_all_results.csv. This file contains the pairwise mutual information between the eleven high-accuracy methods evaluated against the WHO Tumor Classification System all editions.

Documents

Supplementary Document SD1: SD1_Condition_MeSH_Term_Comparison.pdf. This file compares the MeSH terms and conditions terms in the CTR and establishes that even though MeSH terms are standardized, they are not an appropriate representation of the condition names.

Supplementary Document SD2: SD2_Intervention_Type_NIH_CTR.pdf. This file lists the different types of intervention types found in the CTR.

Supplementary Document SD3: SD3_Tumor_Key_Words.pdf. This file contains the tumor key words that were used to detect tumor from CTR, along with standardized terms in ST3 and ST11.

Supplementary Document SD4: SD4_NIH_CTR_Conditions_Discrepancies.pdf . This file displays examples of condition names from the CTR and the various discrepancies associated with them and if there are any standardized terms from the WHO Tumor Classification System.

Supplementary Document SD5: SD5_Example_Edit_Distance.pdf. This file demonstrates an example of how edit distance can be used to transform one string to another.

Supplementary Document SD6: SD6_data_download_instructions.pdf. This file shows the steps involved in downloading the CTR database.